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Kwon et al.

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(54) **FOLDED MODULE AND PORTABLE ELECTRONIC DEVICE INCLUDING A FOLDED MODULE**

(58) **Field of Classification Search**

None

See application file for complete search history.

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Primary Examiner — Derek S. Chapel

(74) *Attorney, Agent, or Firm* — NSIP Law

Related U.S. Application Data

(63) Continuation of application No. 17/142,717, filed on Jan. 6, 2021, now Pat. No. 11,656,476.

(57)

ABSTRACT

A folded module is provided. The folded module includes a housing including a first space and a second space, a reflection module disposed in the first space; and a lens module disposed in the second space. A length of the first space in a first axis direction perpendicular to an optical axis direction is longer than a length of the second space in the first axis direction. The housing also includes a carrier, and a rotation holder is disposed in the carrier. At least a portion of the housing, the carrier and the rotation holder is configured to have a rounded shape corresponding to an arc shape of a circle centered on a second axis perpendicular to an optical axis direction and a first axis direction. The first axis direction and the second axis direction are perpendicular to the optical axis direction and cross each other.

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G02B 7/18 (2021.01)

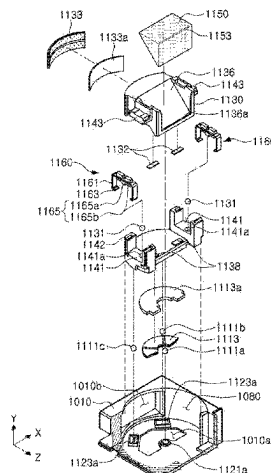
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18 Claims, 25 Drawing Sheets



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G03B 17/17 (2021.01)
H04M 1/02 (2006.01)
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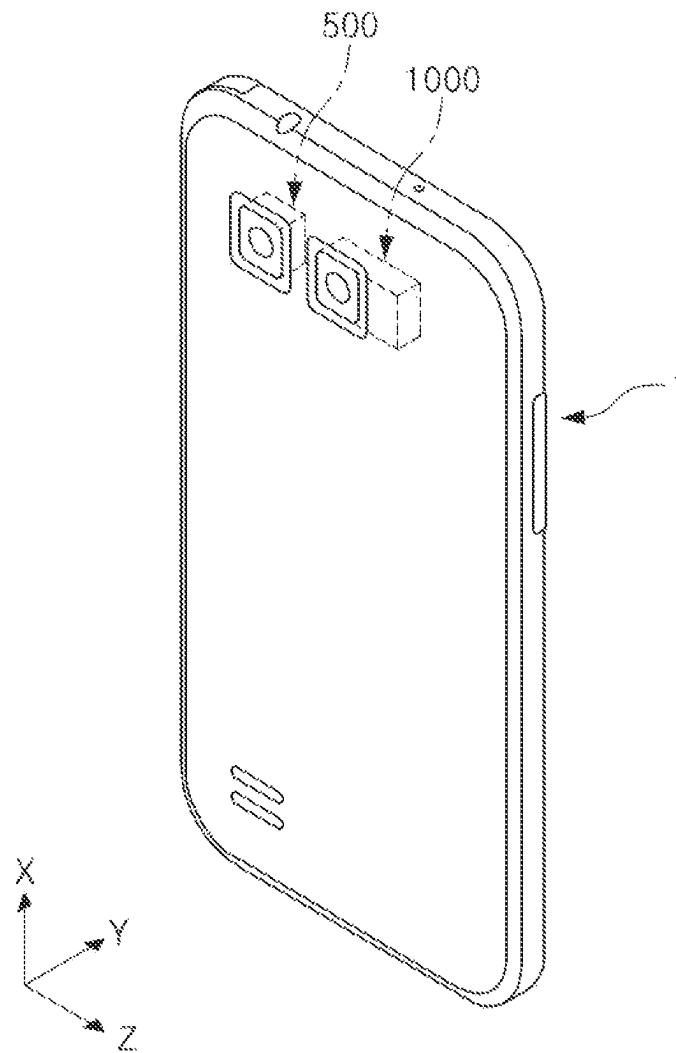


FIG. 1A

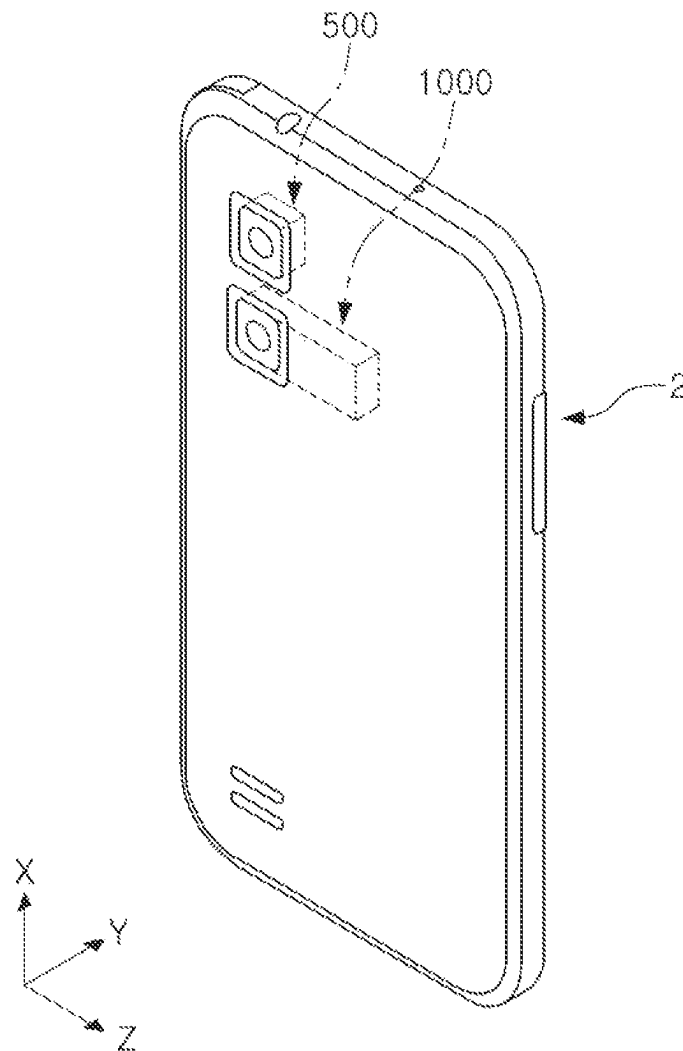


FIG. 1B

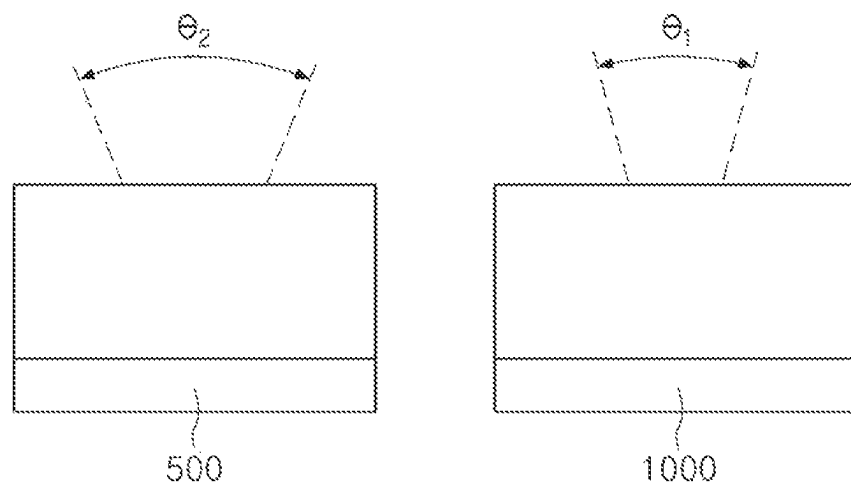


FIG. 2

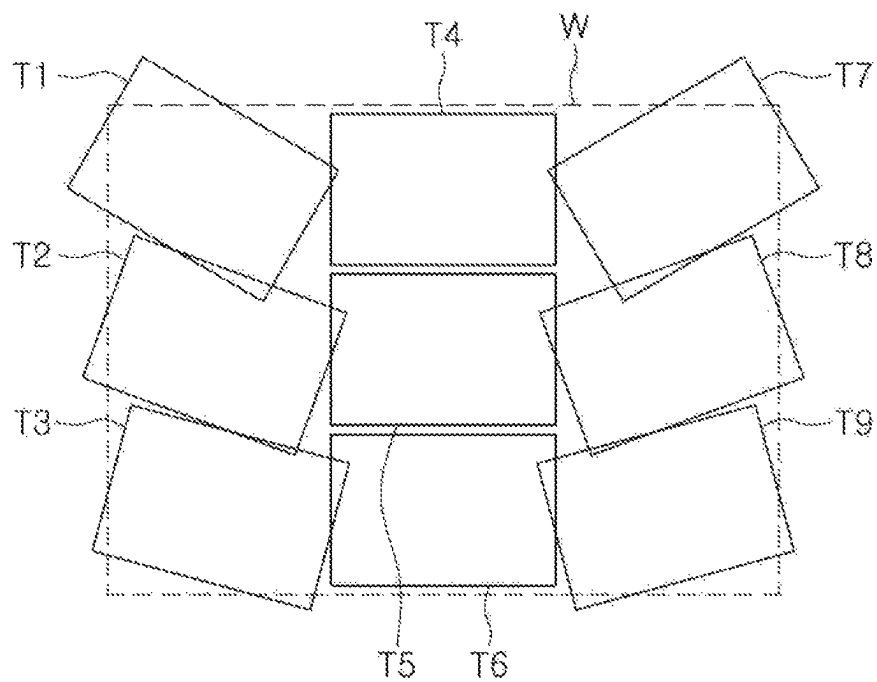


FIG. 3

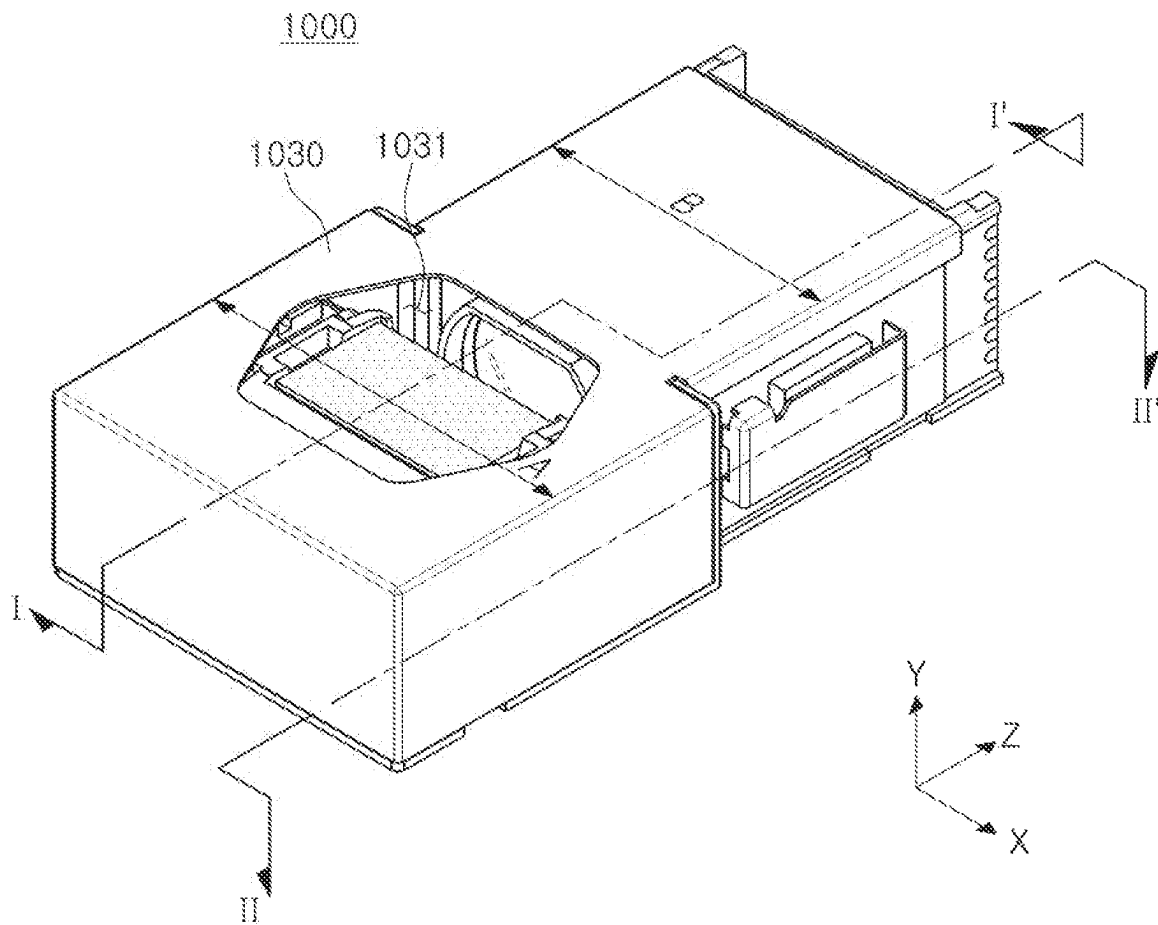


FIG. 4

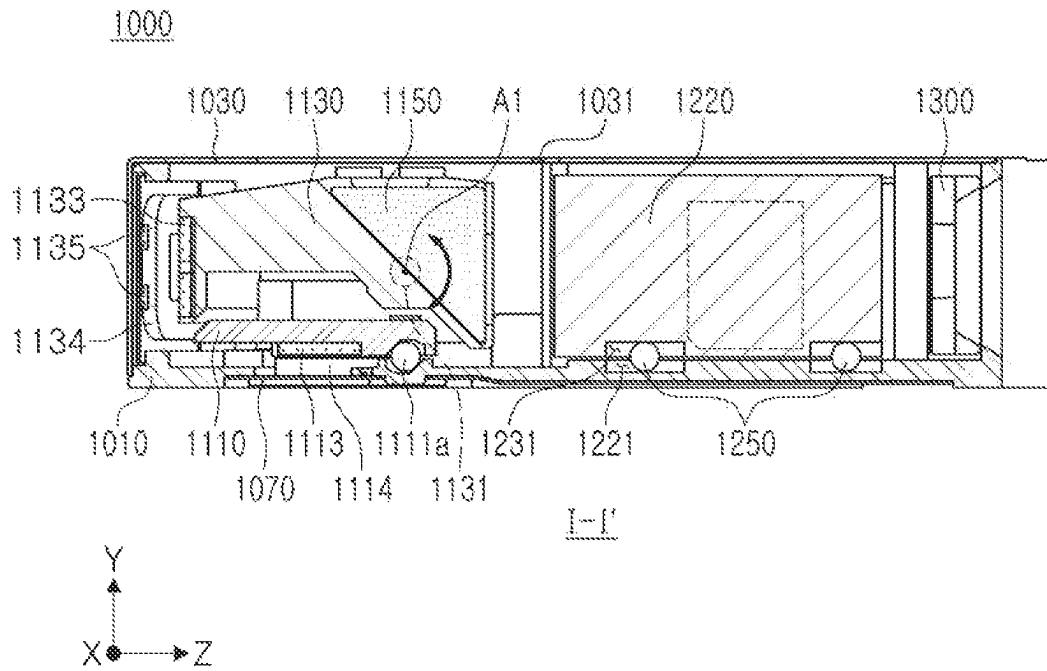


FIG. 5A

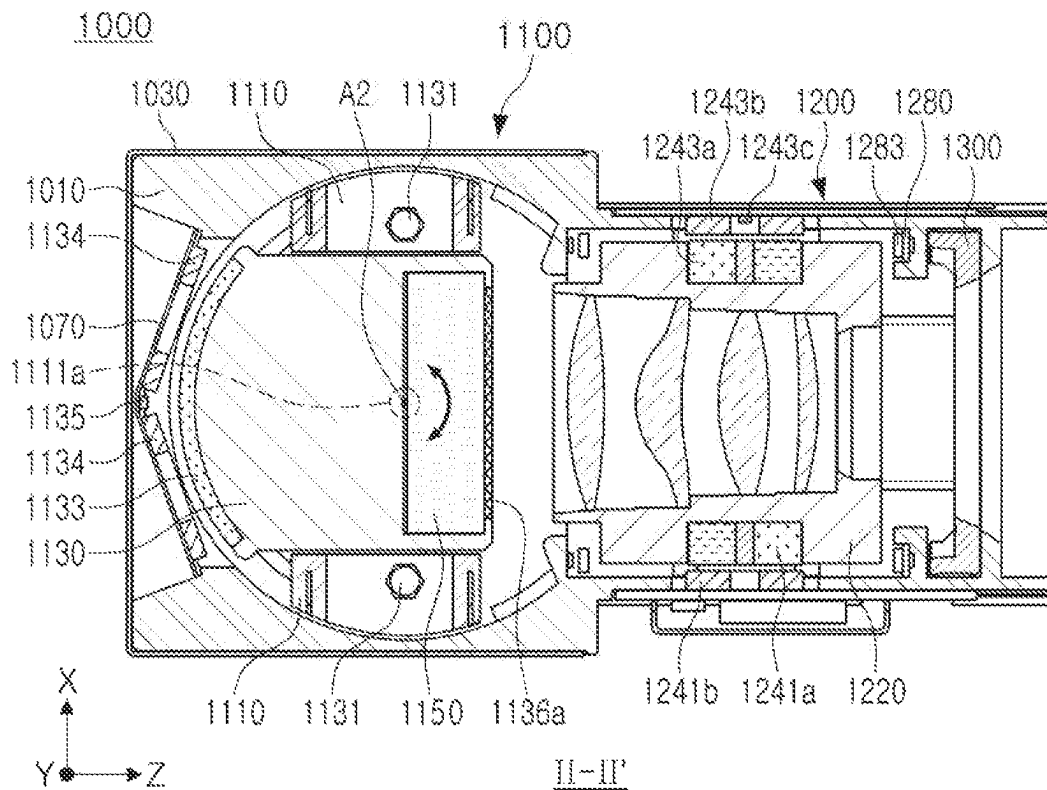


FIG. 5B

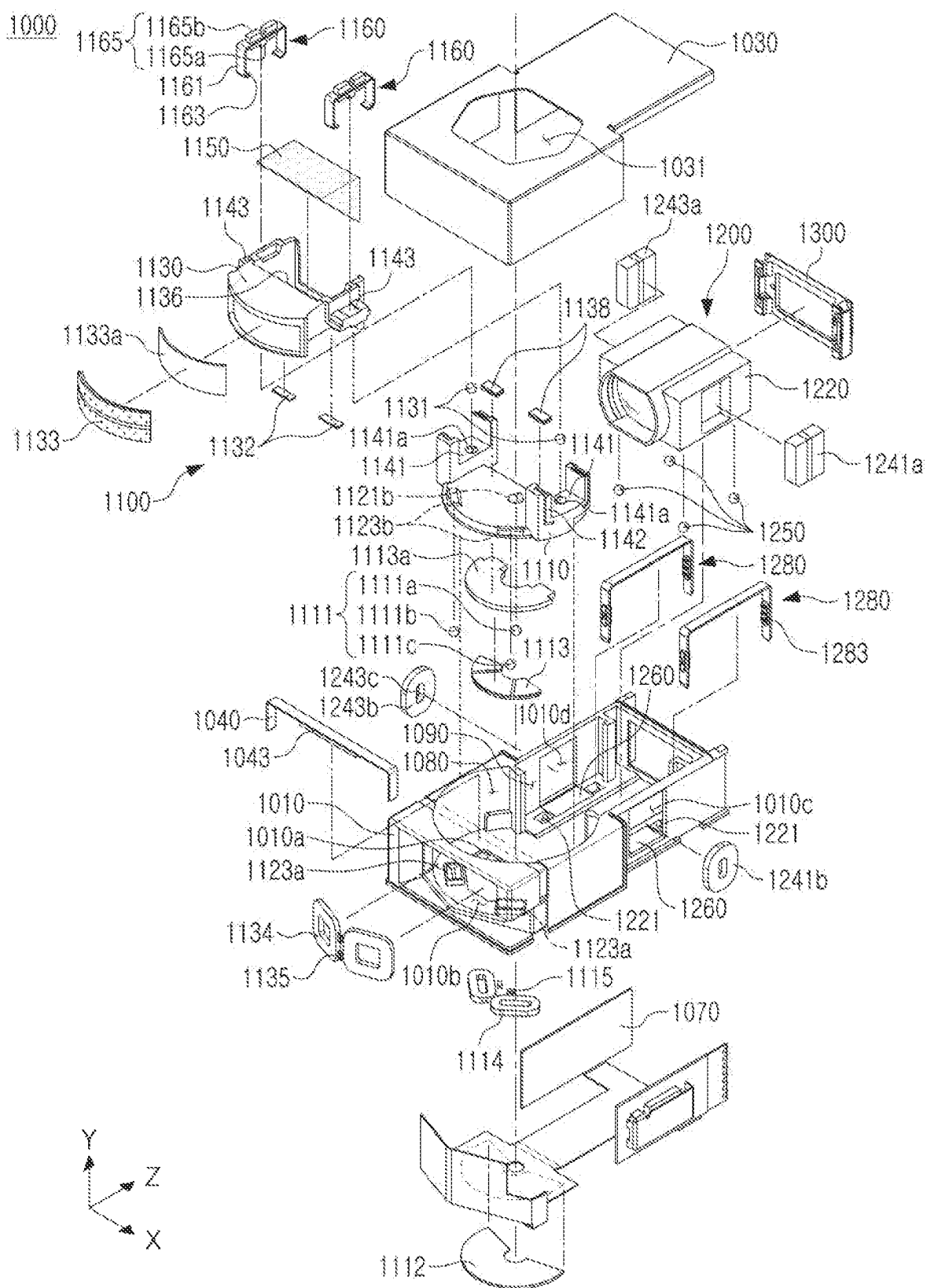


FIG. 6

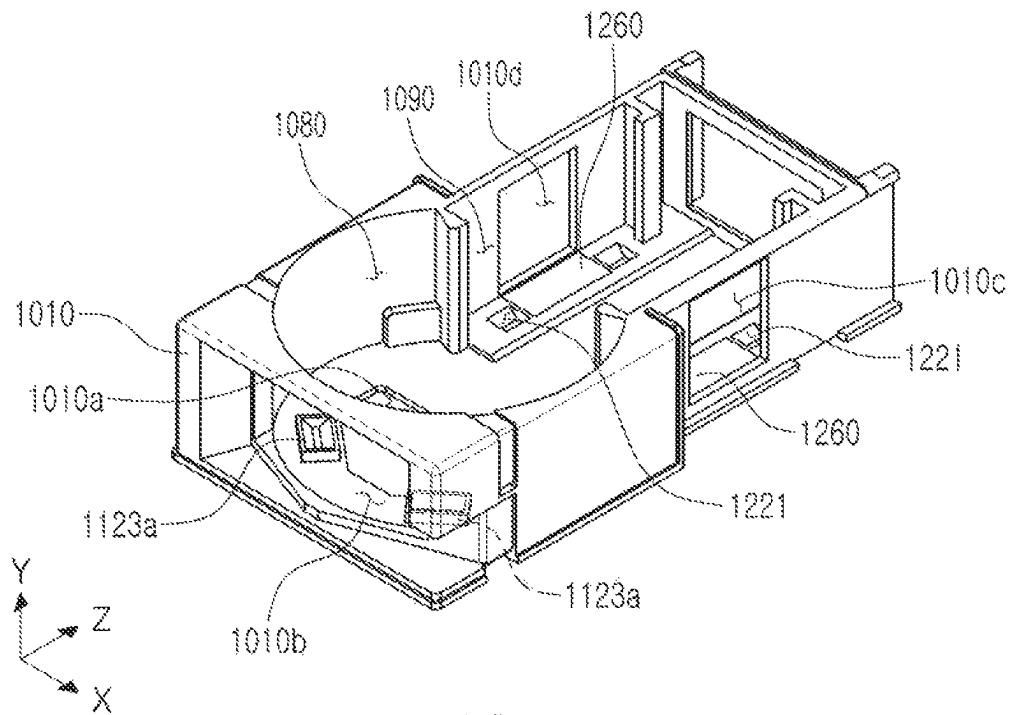


FIG. 7

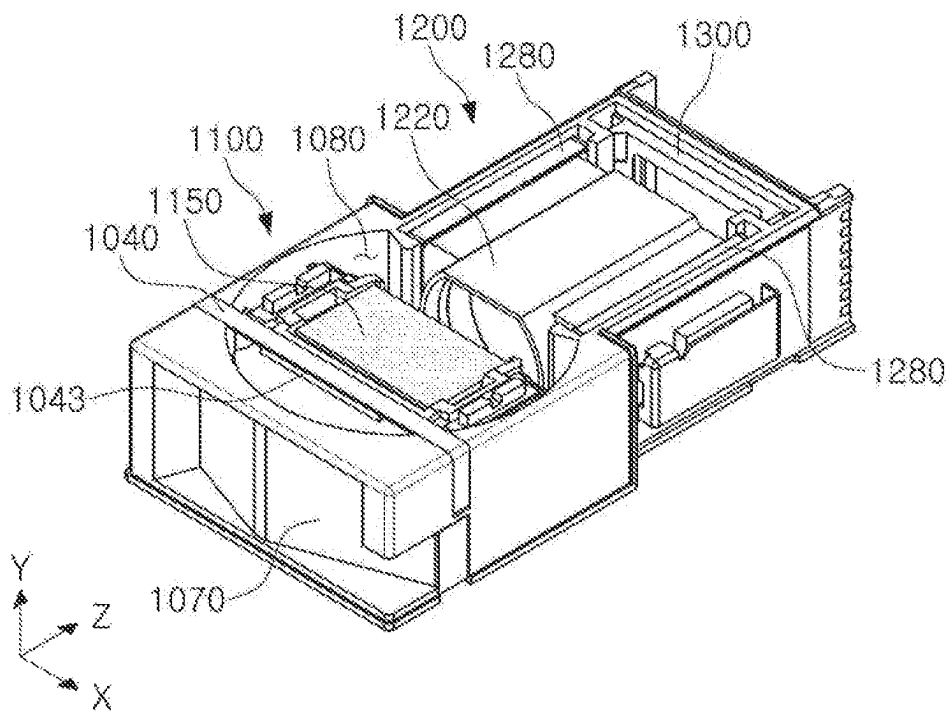


FIG. 8

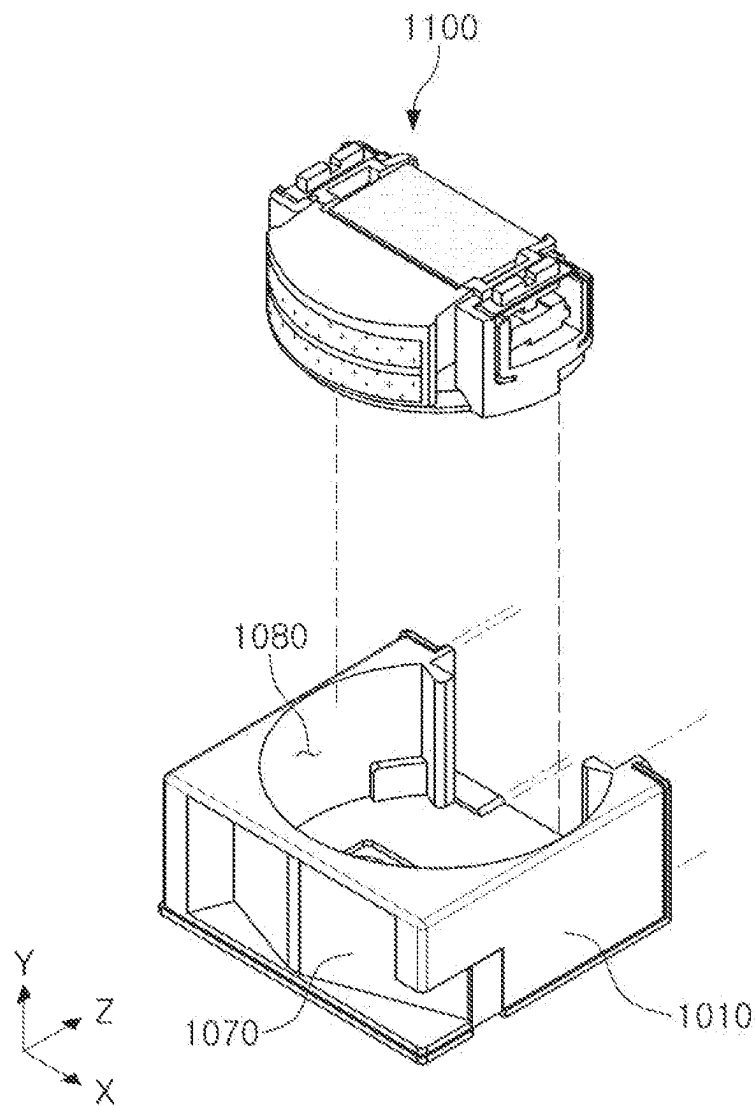


FIG. 9

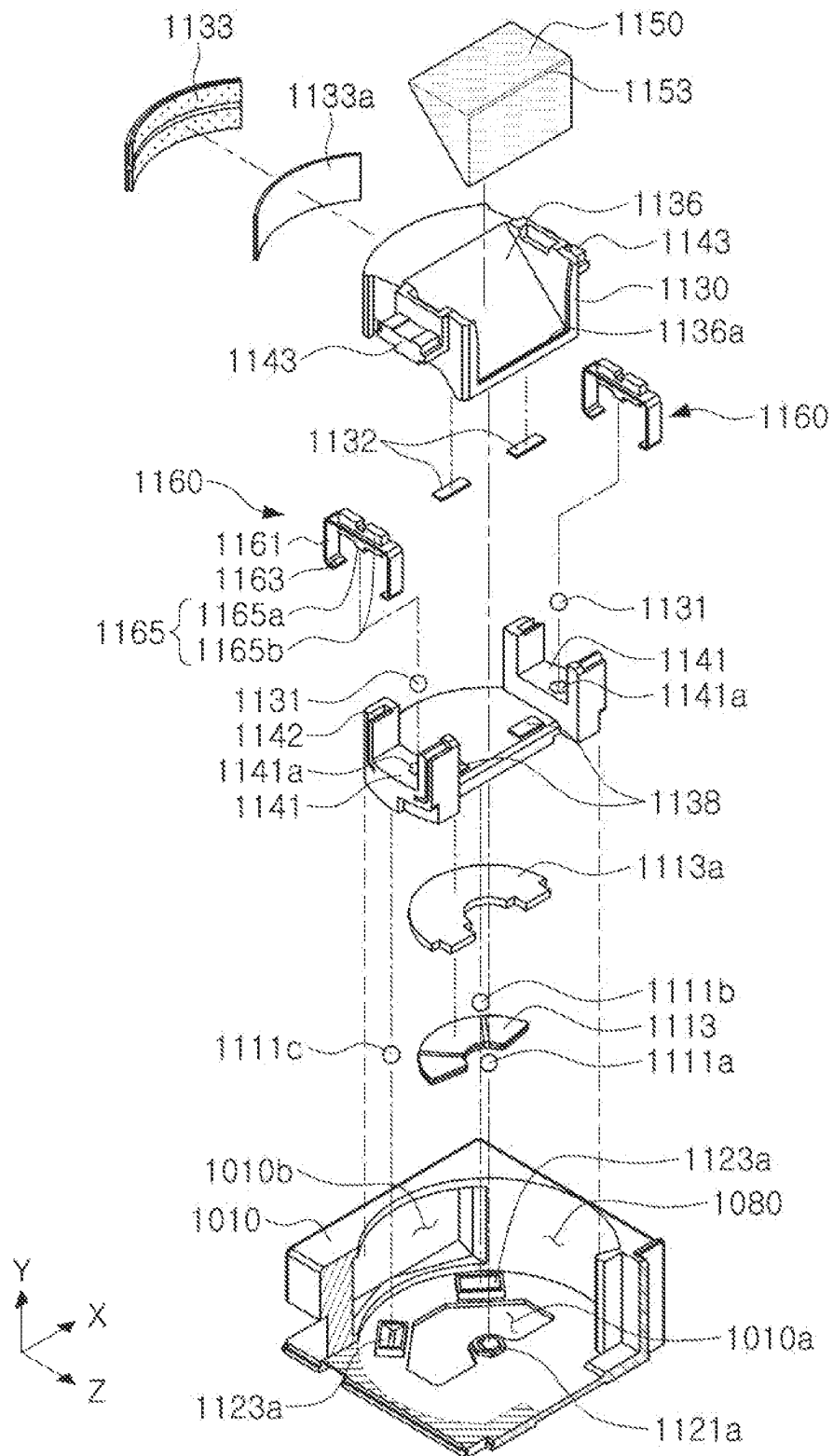


FIG. 10A

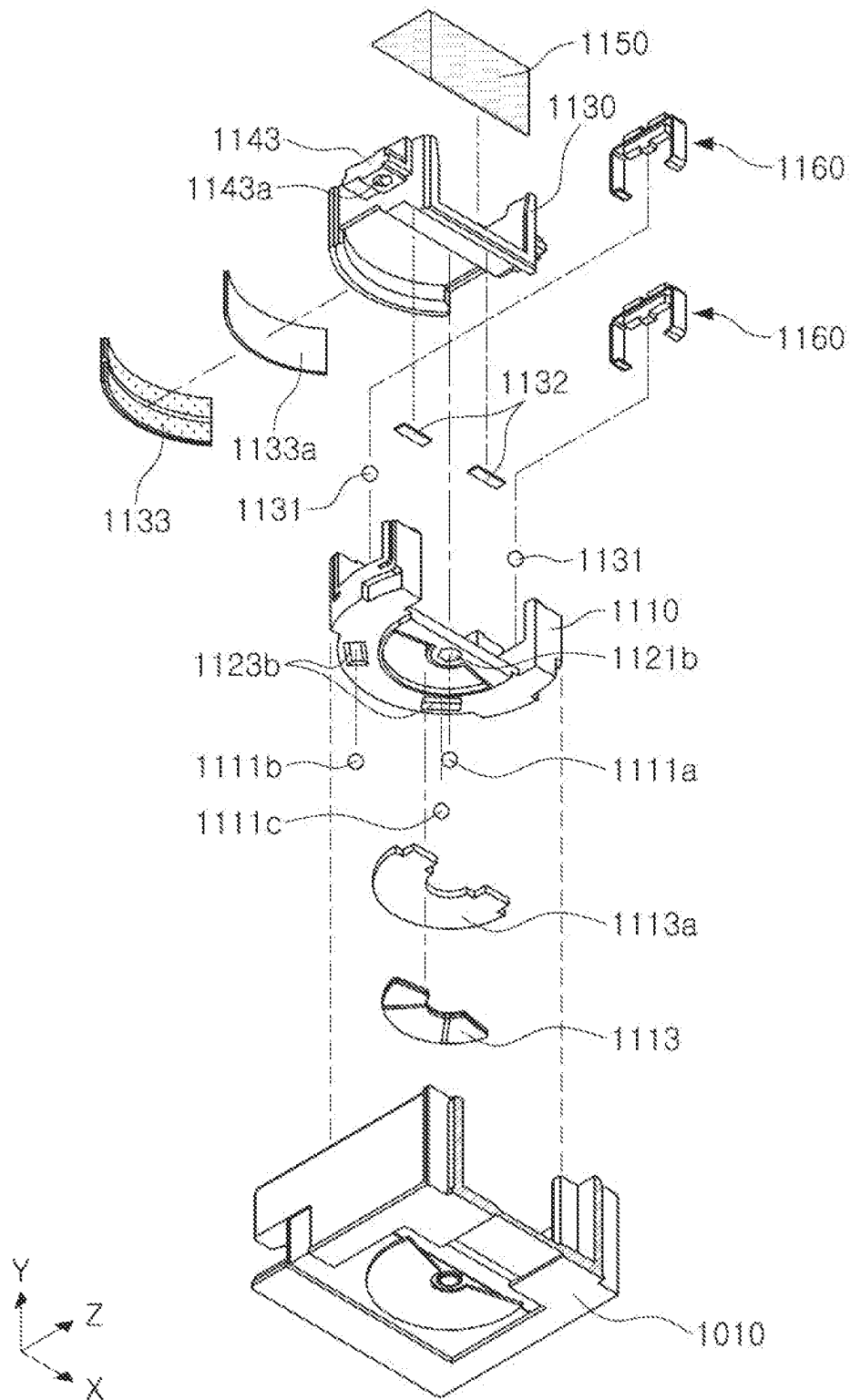


FIG. 10B

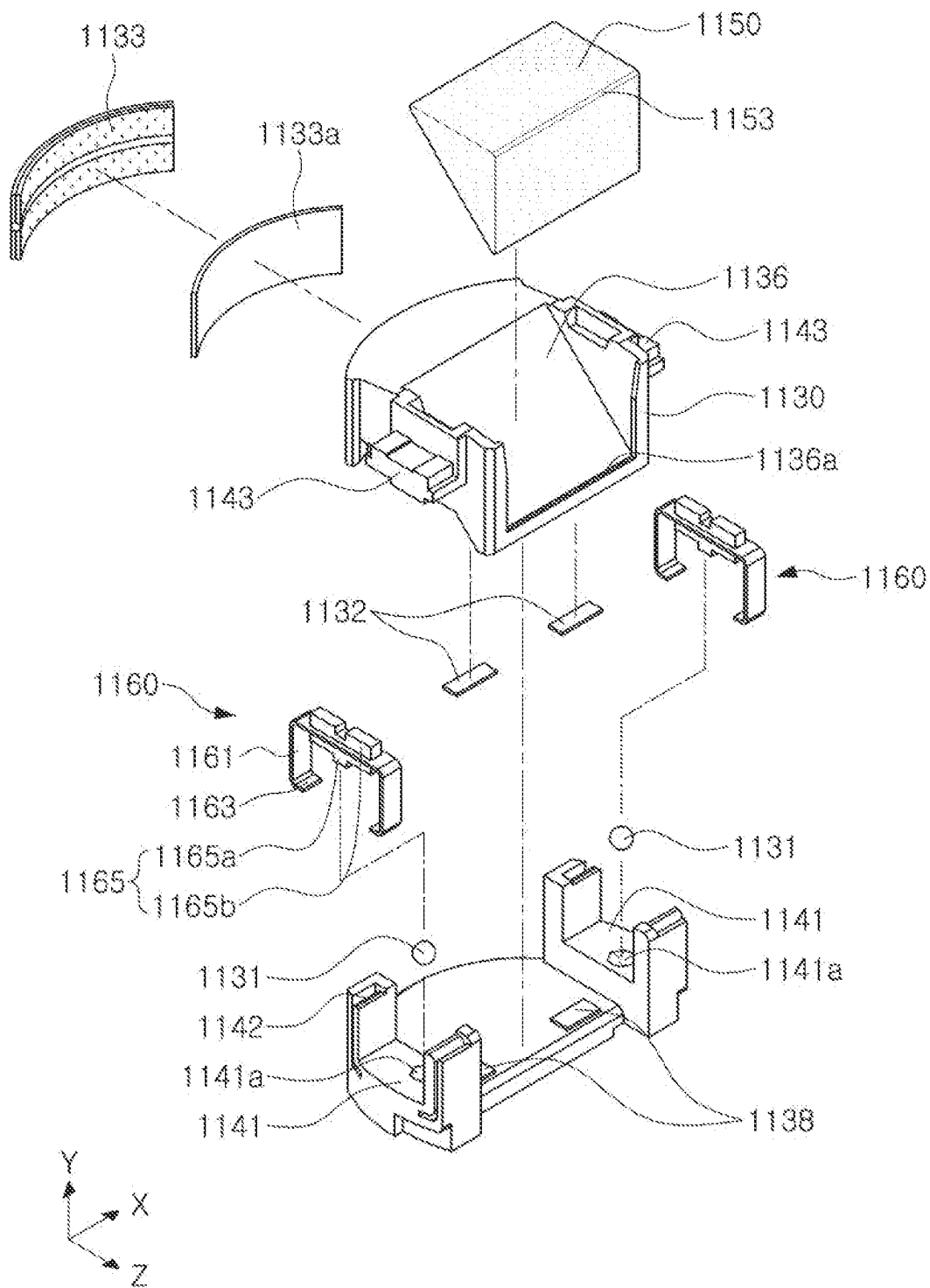


FIG. 11

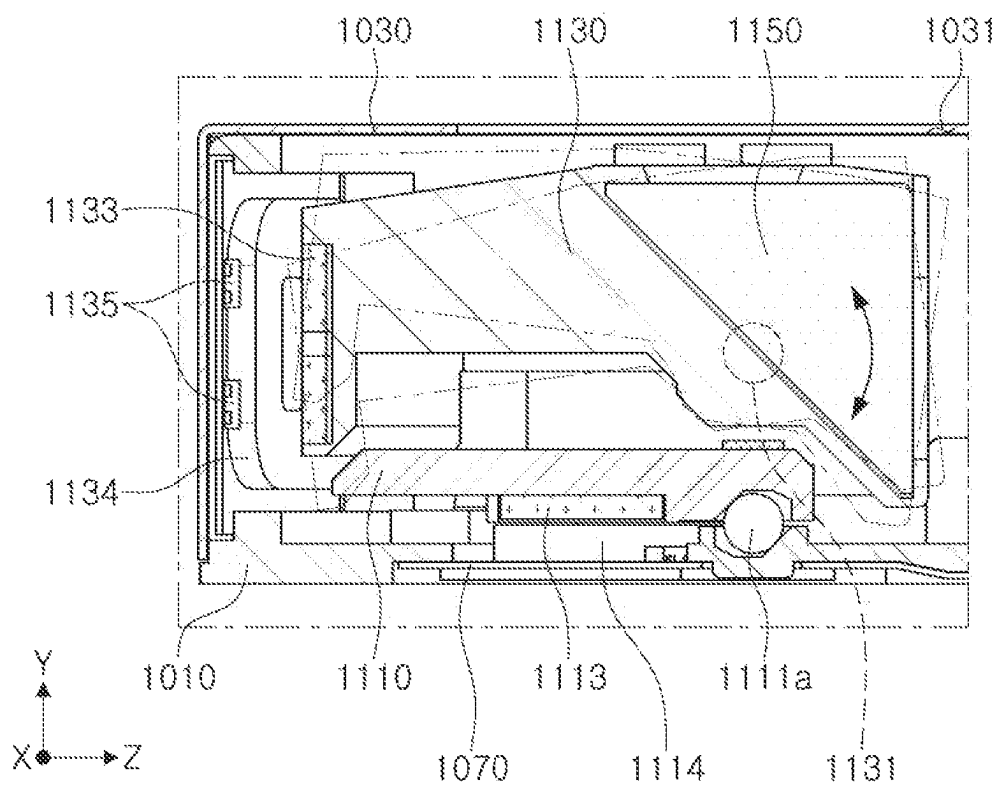


FIG. 13

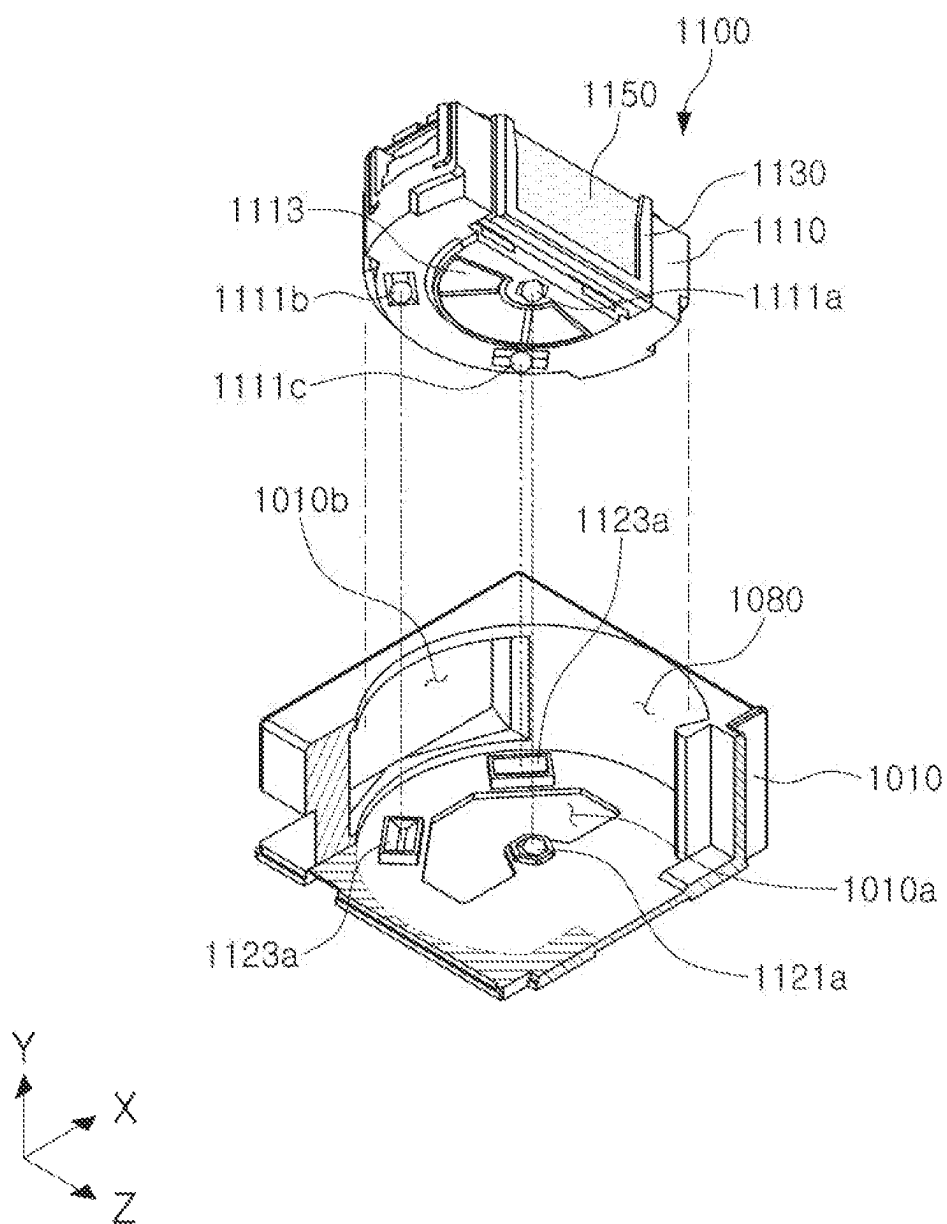


FIG. 14

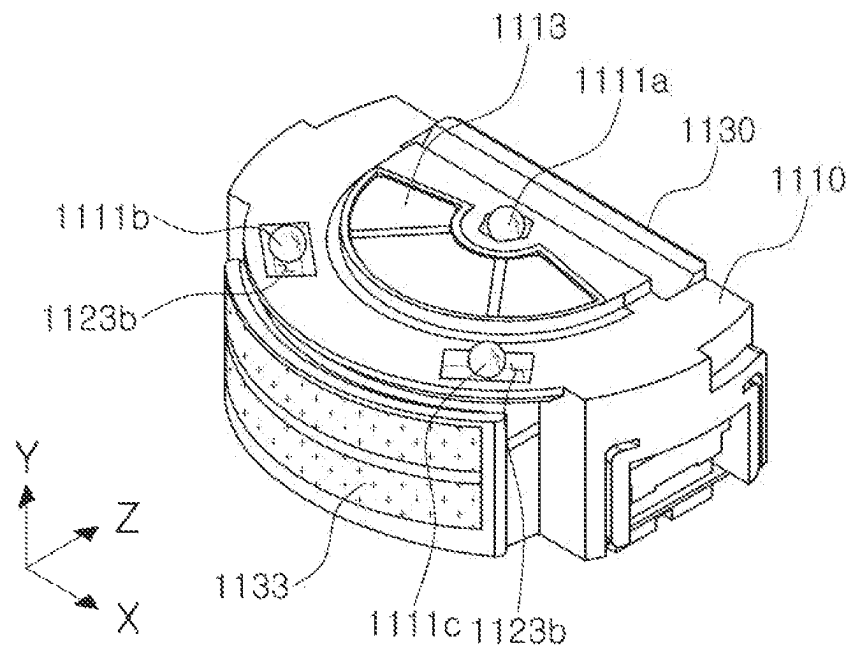


FIG. 15

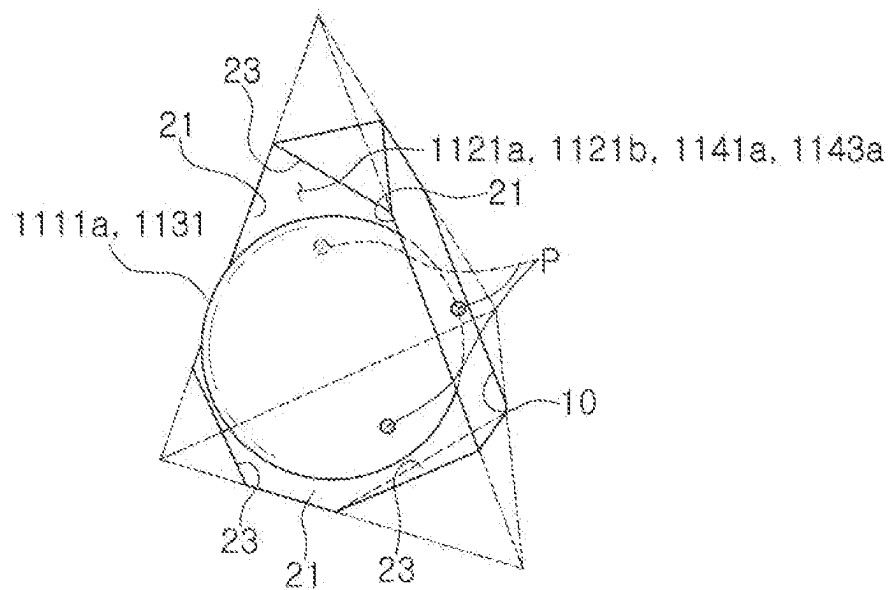


FIG. 16A

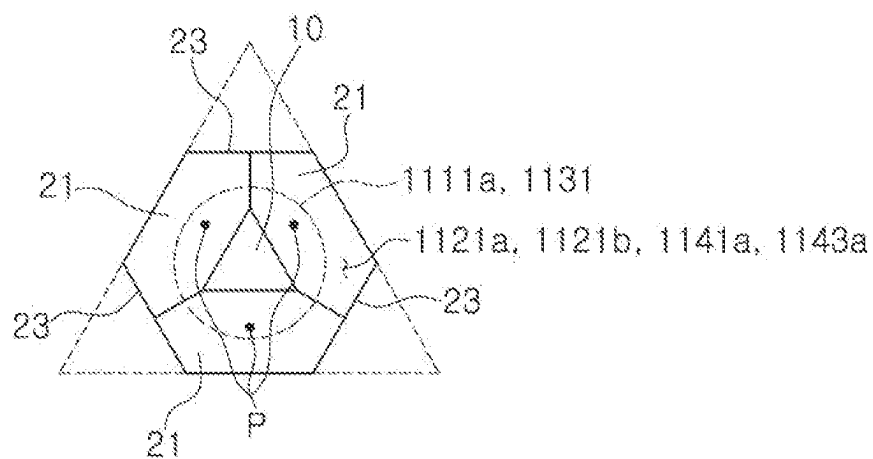


FIG. 16B

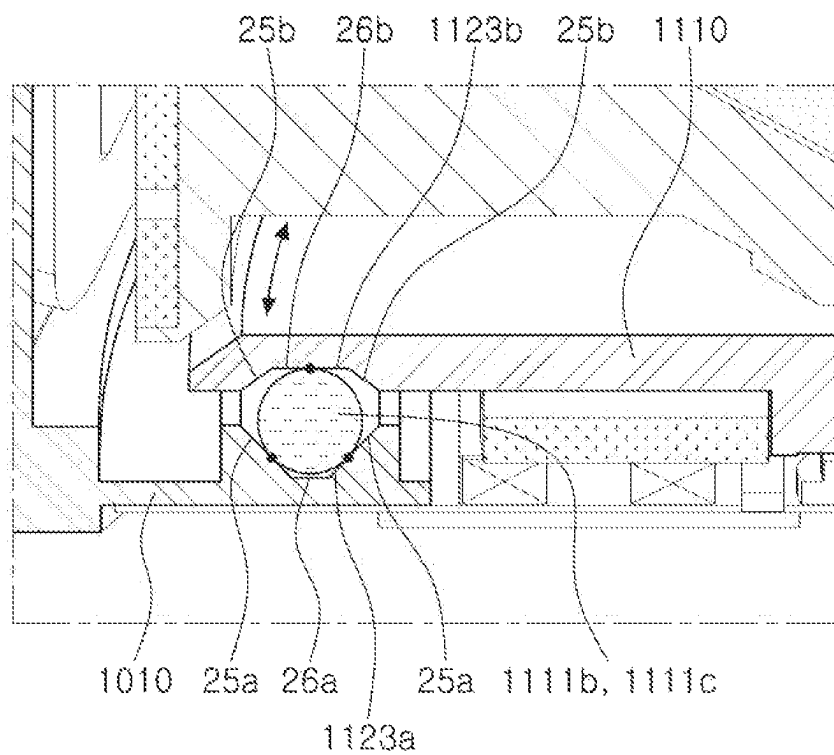


FIG. 17A

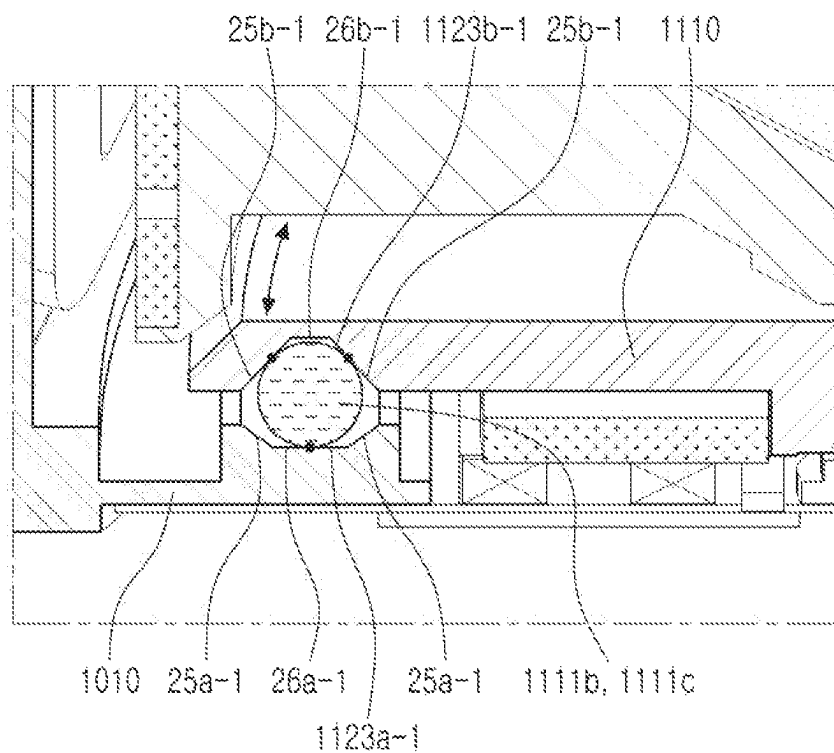


FIG. 17B

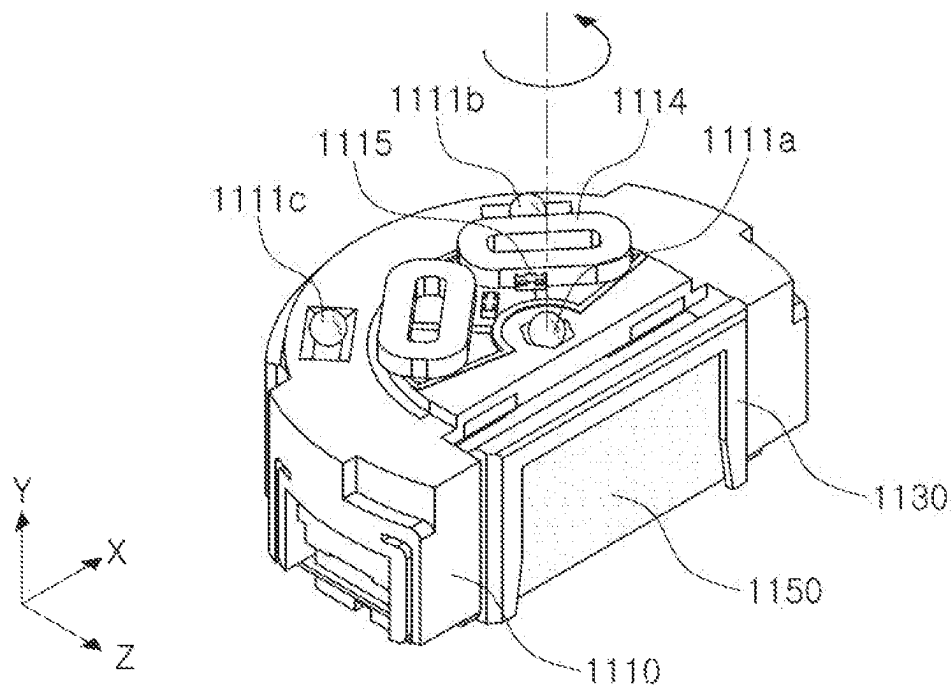


FIG. 18

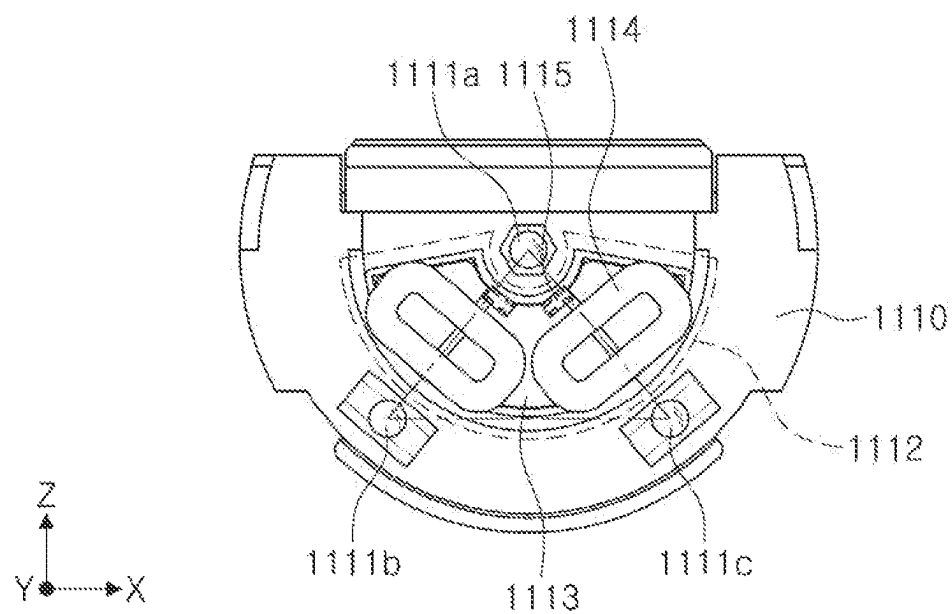


FIG. 19

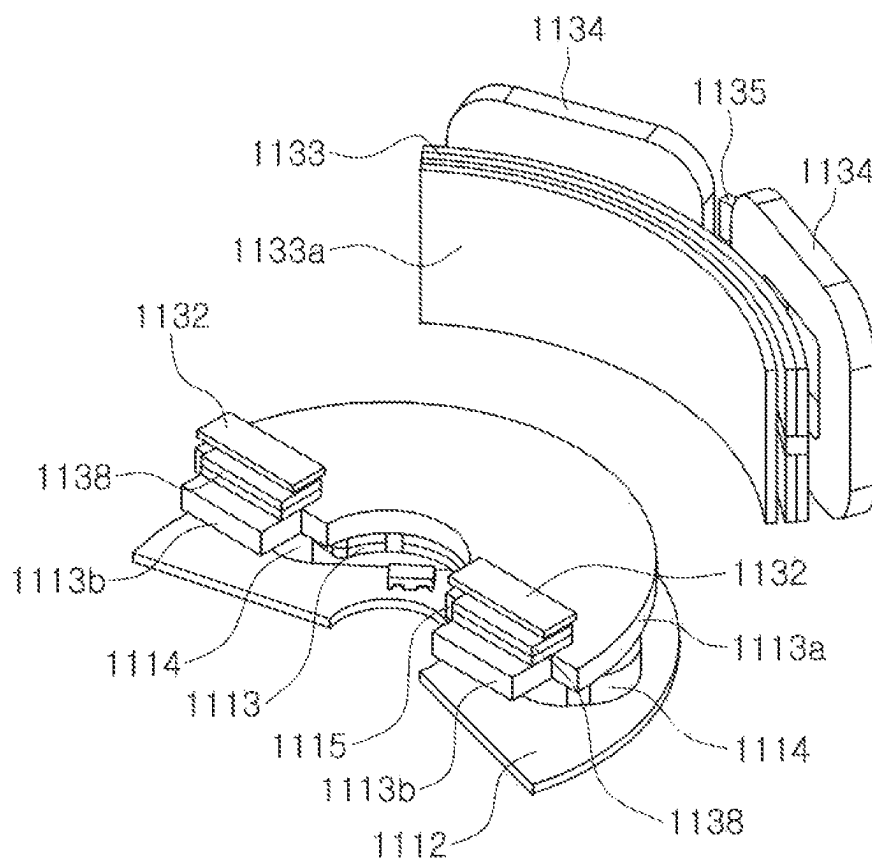


FIG. 20

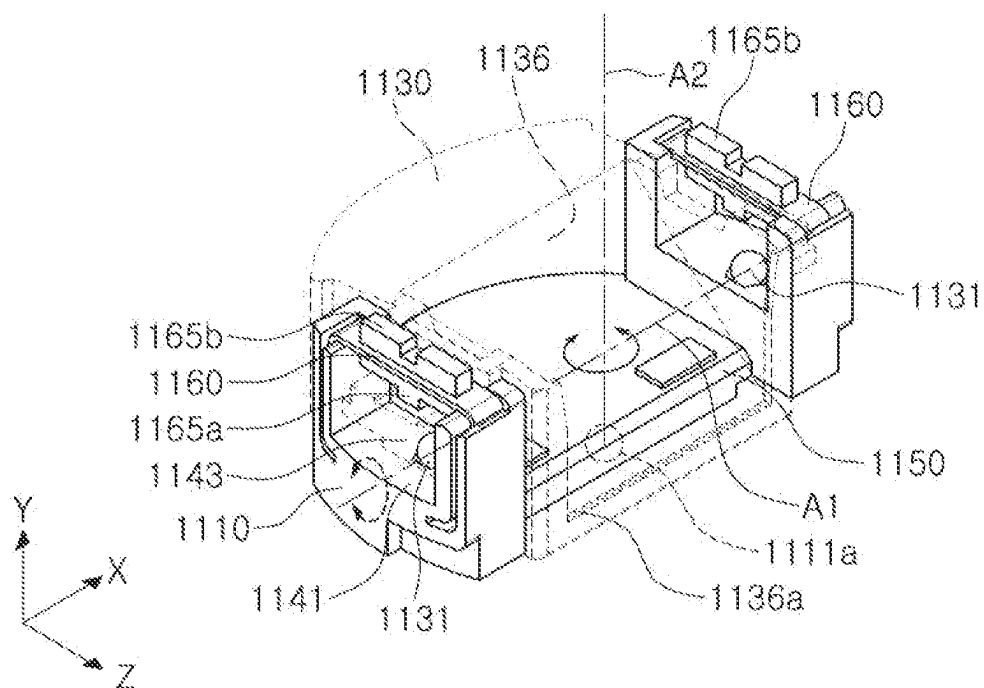


FIG. 21

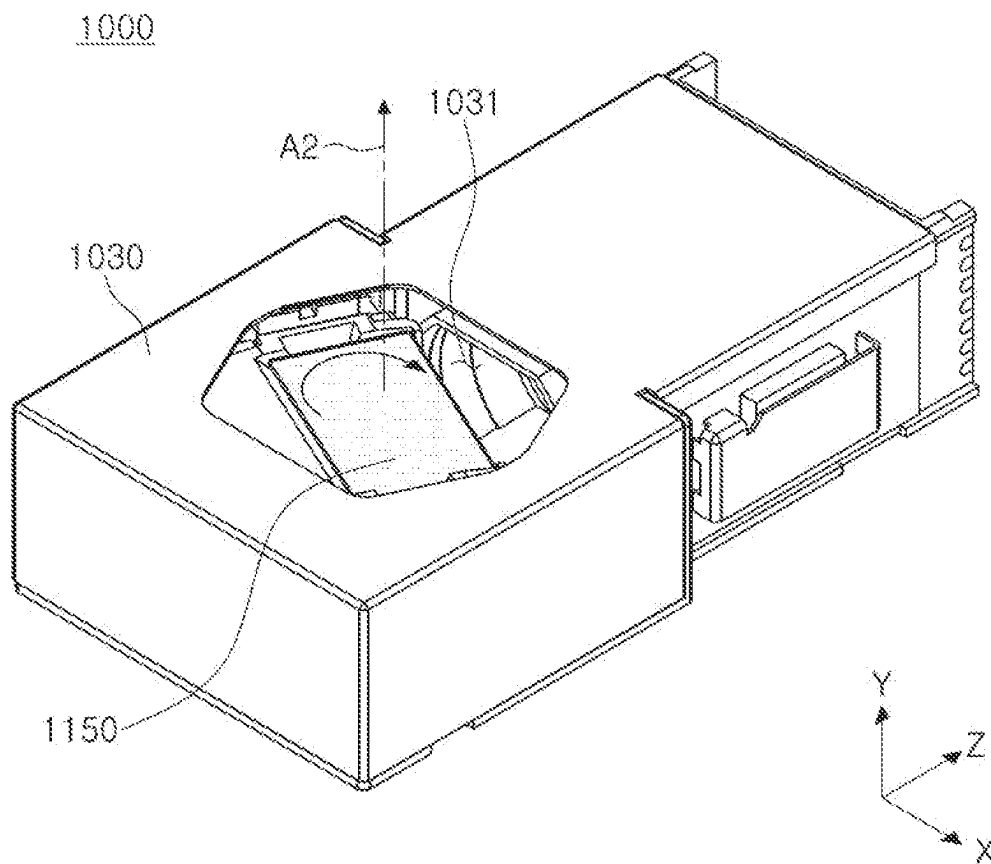


FIG. 22A

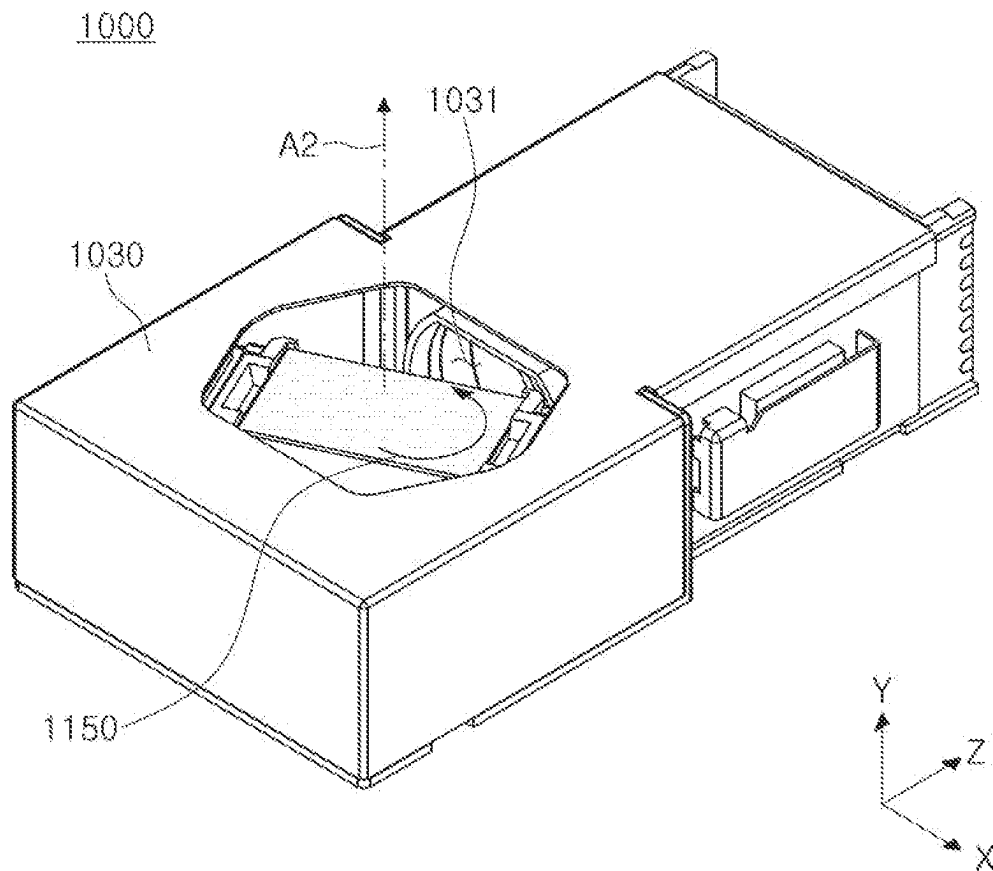


FIG. 22B

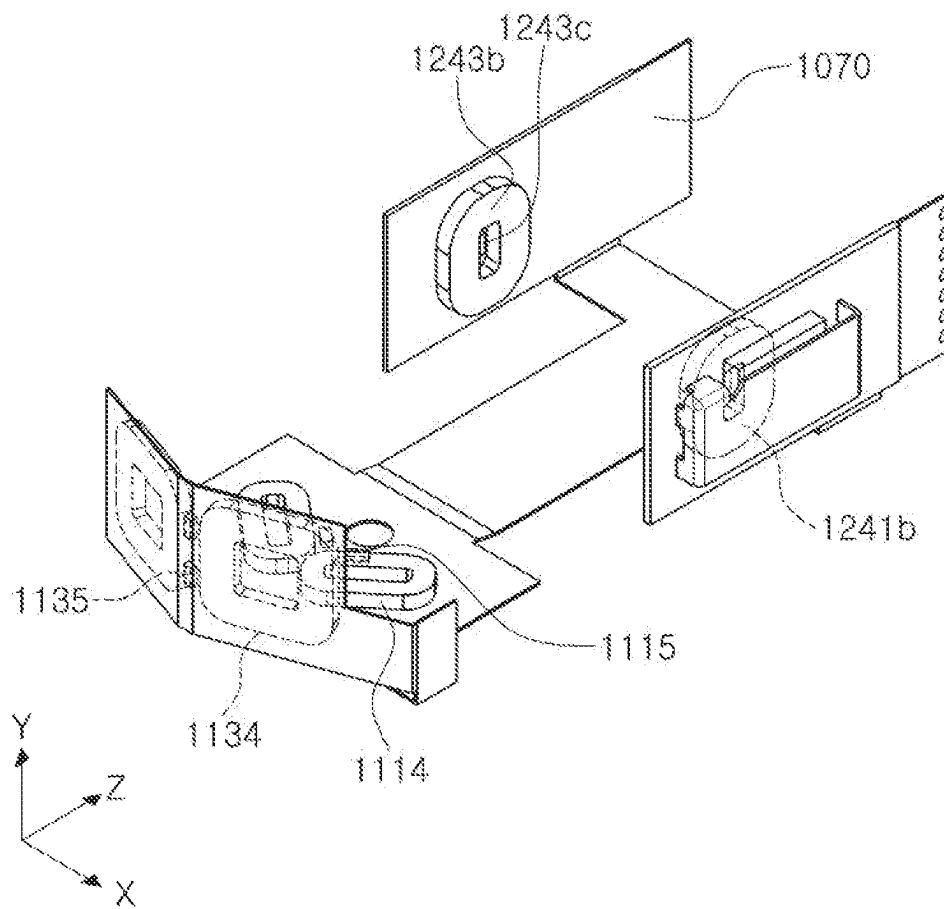


FIG. 23

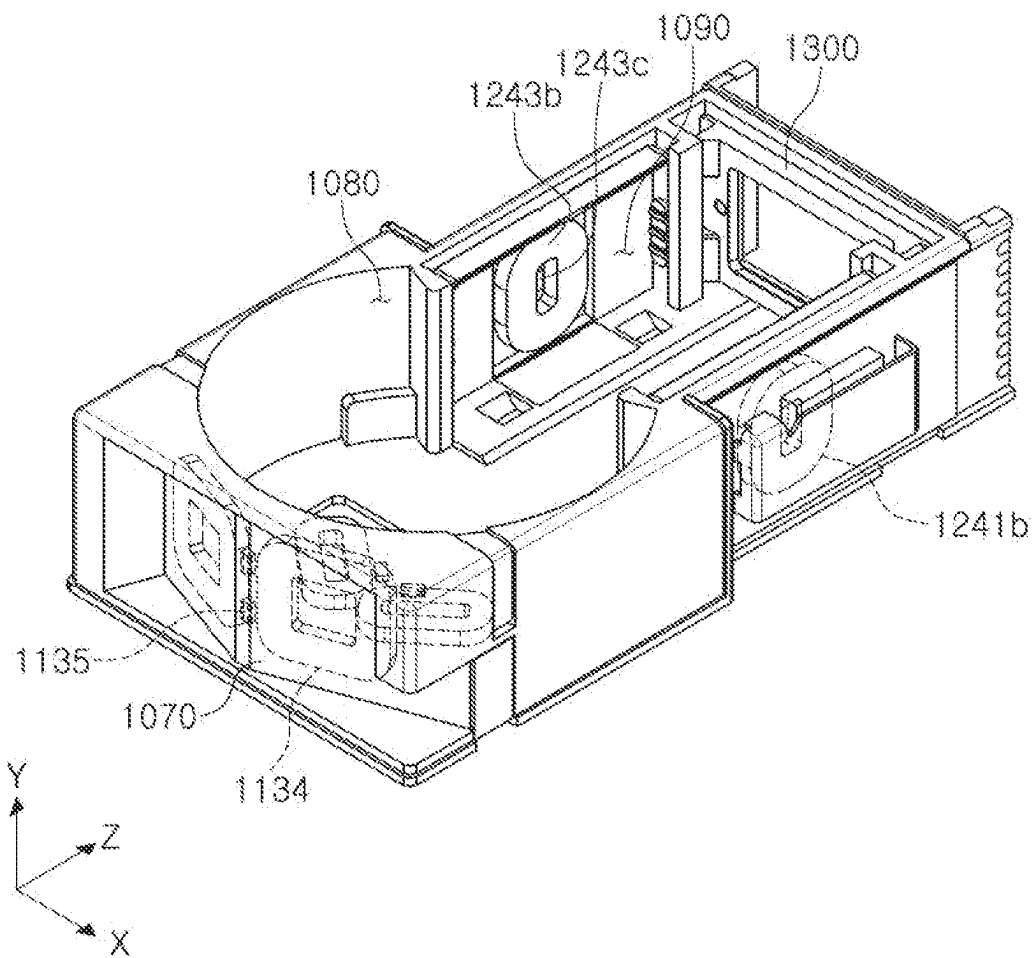


FIG. 24

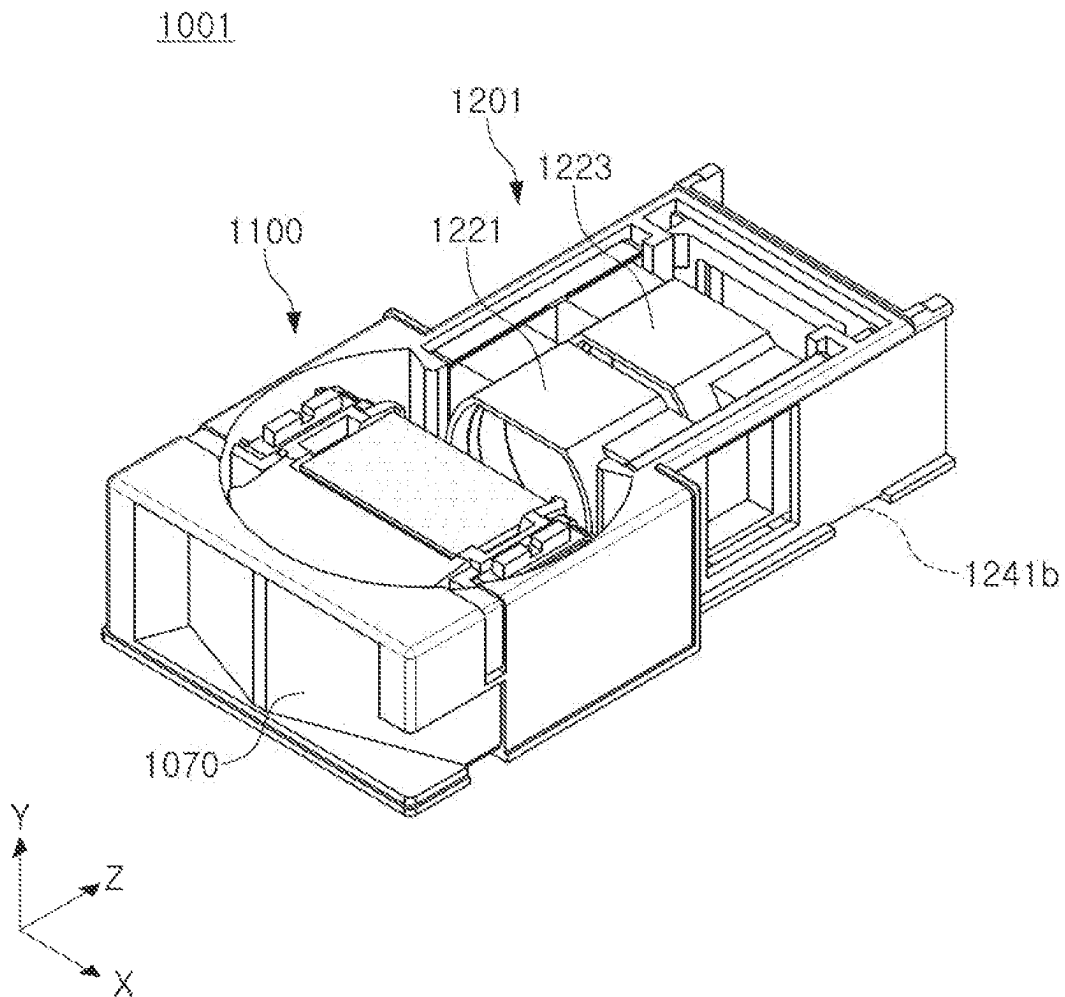


FIG. 25

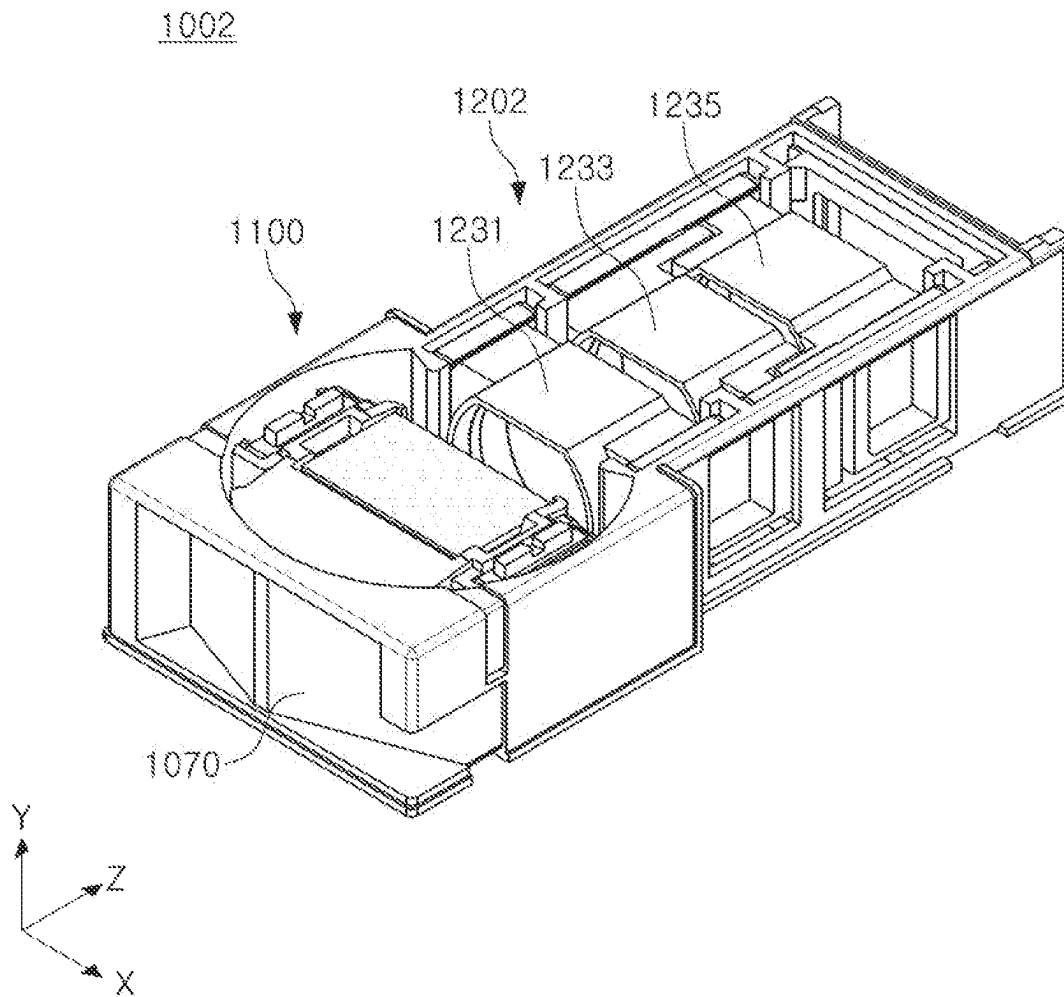


FIG. 26

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FOLDED MODULE AND PORTABLE ELECTRONIC DEVICE INCLUDING A FOLDED MODULE

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of application Ser. No. 17/142,717 filed on Jan. 6, 2021, now U.S. Pat. No. 11,656,476 issued on May 23, 2023, and claims the benefit under 35 USC 119 (a) of Korean Patent Application Nos. 10-2020-0032090 filed on Mar. 16, 2020, and 10-2020-0079633 filed on Jun. 29, 2020, in the Korean Intellectual Property Office, the entire disclosures of which are incorporated herein by reference for all purposes.

BACKGROUND

1. Field

The following description relates to a folded module and a portable electronic device including a Folded Module.

2. Description of the Background

In recent years, cameras have been basically employed in portable electronic devices such as smartphones, tablet personal computers (PCs), and laptops, and autofocus (AF), optical image stabilization (OIS), zoom functions and the like have been added to cameras for mobile terminals.

In addition, a camera module may be provided with an actuator directly moving a lens module or indirectly moving a reflection module including a reflective member to correct shake. In addition, the actuator may move the lens module or the reflection module in a direction intersecting that of an optical axis with driving force of a magnet and a coil.

On the other hand, in recent years, demand for video recording devices has increased rapidly, and there is a problem in that it is difficult to accurately correct shaking when shaking occurs continuously, such as in video recording, in the related art.

In addition, when a subject to be imaged moves during video recording, there may be inconvenience in that the user must directly move a mobile communications terminal to adjust the imaging direction of the camera module to the moving subject, and it may be difficult to accurately capture video footage.

The above information is presented as background information only to assist with an understanding of the present disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

This Summary is provided to introduce a selection of concepts in simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

In one general aspect, a folded module includes a housing, a carrier disposed in the housing, and a rotation holder disposed on the carrier and including a reflective member. The carrier is rotatable, with respect to the housing, around a first axis formed by one rotating shaft ball, the rotation holder is rotatable with respect to the carrier around a second

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axis formed by two ball members, and the first axis and the second axis intersect each other, and the one rotating shaft ball and the two ball members are disposed on a plane including both the first axis and the second axis.

When viewed in a first axis direction, the rotating shaft ball and the two ball members may be aligned in a second axis direction.

The carrier may include a first magnet for driving, and the rotation holder may include a second magnet for driving, and the first magnet and the second magnet may have a rounded shape.

Inner side ends of the first magnet and the second magnet may correspond to an arc shape of a circle centered on the rotating shaft ball.

The housing may include a first coil and a second coil which face the first magnet and the second magnet, in the housing.

At least one of the first coil and the second coil may be provided in plural.

The at least one of the first coil and the second coil provided in plural may be disposed in a bent shape as a whole.

The carrier may be supported on the housing by the one rotating shaft ball and two guide balls.

The carrier may include a first magnet for driving, and a center of the first magnet may be disposed inside of a triangle formed by the one rotating shaft ball and the two guide balls.

The guide ball may be disposed in a first guide portion and a second guide portion which are provided in the carrier and the housing, respectively.

The first guide portion and the second guide portion may have a linear shape.

One of the first guide portion and the second guide portion may have a width wider than a width of the other, for the guide ball to only contact a bottom surface while the guide ball is moving.

The guide ball may be supported by two points on both sides of one of the first guide portion and the second guide portion, and may be supported by one point only on a bottom surface of the other.

Positions of the one rotating shaft ball and the two ball members may be fixed.

A portable electronic device may include a plurality of camera modules having different angles of view, wherein at least one of the plurality of camera modules may be a camera module comprising the folded module.

The plurality of camera modules may be disposed in a plurality of cameras.

In another general aspect, a folded module includes a housing having an internal space of which at least a portion is rounded, a carrier disposed in the internal space and at least partially rounded, and a rotation holder disposed on the carrier. The carrier is rotatable, with respect to the housing, about a first axis, and both the rounded shape of the internal space and the rounded shape of the carrier correspond to a shape of an arc of a circle centered on the first axis.

A portable electronic device may include a plurality of cameras having camera modules having different angles of view, wherein at least one of the plurality of camera modules may include the folded module.

Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

BRIEF DESCRIPTION OF DRAWINGS

FIGS. 1A and 1B are perspective views of portable electronic devices according to one or more examples.

FIG. 2 is a reference diagram illustrating an imaging angle of view of a plurality of camera modules mounted in a portable electronic device according to one or more examples.

FIG. 3 is a reference diagram illustrating an image display screen of a plurality of camera modules mounted in a portable electronic device according to one or more examples.

FIG. 4 is a perspective view of a camera module according to one or more examples.

FIGS. 5A and 5B are cross-sectional views of a camera module according to one or more examples.

FIG. 6 is an exploded perspective view of a camera module according to one or more examples.

FIG. 7 is a perspective view of a housing of a camera module according to one or more examples.

FIG. 8 is a perspective view of a reflection module and a lens module coupled to a housing of a camera module according to one or more examples.

FIG. 9 is an exploded perspective view of a housing and a reflection module of a camera module according to one or more examples.

FIG. 10A is a detailed upper exploded perspective view of a housing and a reflection module of a camera module according to one or more examples.

FIG. 10B is a detailed lower exploded perspective view of a housing and a reflection module of a camera module according to one or more examples.

FIG. 11 is an exploded perspective view of a holder and a carrier of a camera module according to one or more examples.

FIGS. 12A and 12B are perspective views illustrating shapes in which an auxiliary member (stopper or damper) is coupled to a carrier in a camera module according to one or more examples.

FIG. 13 is a cross-sectional view for schematically illustrating a state in which a holder rotates relative to a carrier in a camera module according to one or more examples.

FIG. 14 is an exploded perspective view of a carrier and a housing of a camera module according to one or more examples.

FIG. 15 is a bottom perspective view of a reflection module of a camera module according to one or more examples.

FIGS. 16A and 16B are reference diagrams illustrating an example in which a ball member of a camera module, according to one or more examples, is fixed to a guide portion to be supported by three points.

FIGS. 17A and 17B are excerpt cross-sectional views of a state in which a carrier is coupled to a housing of a camera module according to one or more examples.

FIG. 18 is a bottom perspective view for schematically illustrating a state in which a carrier rotates relative to a housing in a camera module according to one or more examples.

FIG. 19 is a reference diagram for explaining the arrangement of a ball member in which a carrier is supported by a housing, and a positional relationship thereof with another member, in a camera module, according to one or more examples.

FIG. 20 is a reference diagram illustrating a driving unit used for a reflection module in a camera module according to one or more examples.

FIG. 21 is a reference diagram for explaining the positional relationship of two axes in which a reflection module rotates in a camera module according to one or more examples.

FIGS. 22A and 22B are views for explaining the shape of an opening of a cover (cover member) in a camera module according to one or more examples.

FIG. 23 is a perspective view of an integrated substrate installed in a camera module according to one or more examples.

FIG. 24 is a perspective view illustrating a state in which an integrated substrate is installed in a housing of a camera module according to one or more examples.

FIG. 25 is a reference diagram illustrating that a camera module has two lens barrels according to one or more other examples.

FIG. 26 is a reference diagram illustrating that a camera module has three lens barrels according to one or more still other examples.

Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

Hereinafter, while examples of the present disclosure will be described in detail with reference to the accompanying drawings, it is noted that examples are not limited to the same.

The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent after an understanding of this disclosure. For example, the sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent after an understanding of this disclosure, with the exception of operations necessarily occurring in a certain order. Also, descriptions of features that would be well known in the art may be omitted for increased clarity and conciseness.

The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided merely to illustrate some of the many possible ways of implementing the methods, apparatuses, and/or systems described herein that will be apparent after an understanding of this disclosure.

Herein, it is noted that use of the term “may” with respect to an embodiment or example, for example, as to what an embodiment or example may include or implement, means that at least one embodiment or example exists in which such a feature is included or implemented while all embodiments and examples are not limited thereto.

Throughout the specification, when an element, such as a layer, region, or substrate, is described as being “on,” “connected to,” or “coupled to” another element, it may be directly “on,” “connected to,” or “coupled to” the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being “directly on,” “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween. As used herein “portion” of an element may include the whole element or less than the whole element.

As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed

items; likewise, “at least one of” includes any one and any combination of any two or more of the associated listed items.

Although terms such as “first,” “second,” and “third” may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

Spatially relative terms such as “above,” “upper,” “below,” “lower,” and the like, may be used herein for ease of description to describe one element’s relationship to another element as illustrated in the figures. Such spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, an element described as being “above” or “upper” relative to another element will then be “below” or “lower” relative to the other element. Thus, the term “above” encompasses both the above and below orientations depending on the spatial orientation of the device. The device may also be oriented in other ways (for example, rotated 90 degrees or at other orientations), and the spatially relative terms used herein are to be interpreted accordingly.

The terminology used herein is for describing various examples only, and is not to be used to limit the disclosure. The articles “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “includes,” and “has” specify the presence of stated features, numbers, operations, members, elements, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, members, elements, and/or combinations thereof.

Due to manufacturing techniques and/or tolerances, variations of the shapes illustrated in the drawings may occur. Thus, the examples described herein are not limited to the specific shapes illustrated in the drawings, but include changes in shape that occur during manufacturing.

The features of the examples described herein may be combined in various ways as will be apparent after gaining an understanding of this disclosure. Further, although the examples described herein have a variety of configurations, other configurations are possible as will be apparent after an understanding of this disclosure.

Examples disclosed herein describe a folded module in which shaking may be easily controlled in not only imaging a fixed subject, but also in capturing video of a moving subject, and a camera module including the same.

Examples disclosed herein describe a folded module capable of tracking a moving subject and correcting shake, and a portable electronic device including the same.

FIGS. 1A and 1B are perspective views of portable electronic devices according to one or more examples, and FIG. 2 is a reference diagram illustrating an imaging angle of view of a plurality of camera modules mounted in a portable electronic device according to one or more examples, and FIG. 3 is a reference diagram illustrating an image display screen of a plurality of camera modules mounted in a portable electronic device according to one or more examples.

Portable electronic devices **1** and **2** may be portable electronic devices such as mobile communication terminals, smart phones, and tablet PCs.

As illustrated in FIGS. 1A and 1B, a plurality of camera modules are mounted in the portable electronic devices **1** and **2** to capture a subject. For example, the portable electronic device may include a first camera module **1000** and a second camera module **500**.

In each of FIGS. 1A and 1B, two camera modules may be provided. FIG. 1A illustrates a case in which the first camera module **1000** and the second camera module **500** are sequentially disposed in a width direction (a relatively short side direction) of the portable electronic device **1**, while FIG. 1B illustrates a case in which the first camera module **1000** and the second camera module **500** are sequentially disposed in a length direction (relatively long side direction) of the portable electronic device **2**.

In the case of using two camera modules, entrance ports through which light is incident on the two camera modules may be disposed to be as close to each other as possible.

Further, as illustrated in FIG. 2, the first camera module **1000** and the second camera module **500** are configured to have different angles of view.

The first camera module **1000** is configured with a relatively narrow angle of view (e.g., a telephoto camera), and the second camera module **500** is configured with a relatively wide angle of view (e.g., a wide-angle camera). In this case, the first camera module **1000** may correspond to a camera module described below with reference to FIGS. 4 to 25.

For example, the angle of view $\theta 1$ of the first camera module **1000** may be formed in the range of 9° to 35° , and the angle of view $\theta 2$ of the second camera module **500** may be formed in the range of 60° to 120° .

By designing different angles of view of the two camera modules as described above, an image of a subject may be imaged at various depths.

On the other hand, the portable electronic devices **1** and **2** according to one or more examples may have a Picture in Picture (PIP) function.

For example, the portable electronic device **1**, **2** may display an image captured by a camera module having a relatively narrower field of view (e.g., by the first camera module **1000**) in an image captured by a camera module having a relatively wider field of view (e.g., by the second camera module **500**).

For example, a subject of interest may be imaged with a narrow angle of view (thus, the subject of interest is enlarged) and may be displayed in an image imaged with a wide angle of view.

When shooting a video, a subject of interest may move, and thus, a camera module having a narrower angle of view (for example, the first camera module **1000**) may have a reflection module (folded module) that rotates to capture an image according to the movement of the subject of interest. Accordingly, the light incident on the first camera module **1000** may be reflected by a reflective member of the reflection module so that the light path is changed and then incident on a lens module.

For example, the first camera module **1000** may rotate the reflection module to track the movement of the subject of interest.

For example, the reflection module provided in the first camera module **1000** may be rotated based on a first axis (X axis) and a second axis (Y axis). Accordingly, the first camera module **1000** may correct shake that may occur during imaging.

In this case, the first axis (X axis) means an axis perpendicular to the optical axis (Z axis), and the second axis (Y axis) means an axis perpendicular to both the optical axis (Z axis) and the first axis (X axis). In addition, the first axis (X axis) and the second axis (Y axis), rotation axes of the reflection module of the first camera module **1000**, may intersect the optical axis (Z axis); and the optical axis (Z axis), the first axis (X axis) and the second axis (Y axis) may meet at approximately any one point.

FIG. 3 illustrates a range of a subject that may be imaged using the first camera module **1000** and the second camera module **500** installed in the portable electronic devices **1** and **2** according to one or more examples.

The second camera module **500** having a relatively wide field of view is capable of imaging a subject of a relatively large area, and the first camera module **1000** having a relatively narrow field of view may image a subject of a relatively narrow area.

In detail, the first camera module **1000** may image an inner area of a wide imaging range W captured by the second camera module **500**, as a teleimaging range T1 to T9, and thus, an image captured as the teleimaging range T1 to T9 may be displayed inside of an image captured as the wide imaging range W. Of course, the teleimaging range T1 to T9 imaged by the first camera module **1000** may overlap a portion of the inner area of the wide imaging range W and an exterior thereof, or may be an outer area of the wide imaging range W.

On the other hand, since the first camera module **1000** includes a reflection module (folded module) that rotates about the first axis (X axis) and the second axis (Y axis) intersecting the optical axis (Z axis), the image captured by the first camera module **1000** may be inclined to an image captured by the second camera module **500** by changing an imaging angle by rotation of the reflection module, which is a case in which the angle of T1 to T3 or T7 to T9 of the teleimaging range illustrated in the reference diagram of FIG. 3 is changed by the rotation of the reflection module.

Accordingly, in the case of an image (video) taken of T1 to T3 or T7 to T9 as a subject, among the teleimaging range taken by the first camera module **1000**, the image may be rotated to be aligned with a captured image (video) of the second camera module **500** to implement the PIP function.

To implement these functions, the camera modules **1000** and **500** of the portable electronic devices **1** and **2** may be provided with a control unit for editing an image or implementing a PIP function.

FIG. 4 is a perspective view of a camera module according to one or more examples, FIGS. 5A and 5B are cross-sectional views of a camera module according to one or more examples, and FIG. 6 is an exploded perspective view of a camera module according to one or more examples, FIG. 7 is a perspective view of a housing of a camera module according to one or more examples, and FIG. 8 is a perspective view in which a reflection module and a lens module are combined in a housing of a camera module according to one or more examples.

Referring to FIGS. 4 to 8, a camera module **1000** according to one or more examples includes a reflection module **1100** (folded module) and a lens module **1200** provided in a housing **1010**.

The reflection module **1100** is configured to change the traveling direction of light. As an example, the travelling direction of the light incident through an opening **1031** of a cover **1030** (for example, a shield can) covering the camera module **1000** from the top may be changed to be directed toward the lens module **1200** through the reflection module

1100. To this end, the reflection module **1100** may include a reflective member **1150** reflecting light.

The reflective member **1150** may include a chamfer **1153** in which a corner of the reflective member **1150** is cut to reduce light reflection or scattering.

The path of light incident in the thickness direction (Y-axis direction) of the camera module **1000** is changed by the reflection module **1100** to substantially coincide with the optical axis (Z-axis) direction.

To this end, the reflection module **1100** includes the reflective member **1150** reflecting light. Then, the light incident on the lens module **1200** may be converted into an electric signal by an image sensor (not illustrated) after passing through the plurality of lenses and then be stored.

The lens module **1200** includes a plurality of lenses through which light whose traveling direction has been changed by the reflection module **1100** passes. In addition, the lens module **1200** includes at least one lens barrel. Autofocusing (AF) or a zoom function may be implemented by the movement of at least one lens barrel in the optical axis direction (Z-axis).

An image sensor module (not illustrated) may include an image sensor (not illustrated) for converting light passing through a plurality of lenses into electric signals, and a printed circuit board (not illustrated) on which the image sensor is mounted.

In an internal space of the housing **1010**, the reflection module **1100** is provided in front of the lens module **1200**, centered on the lens module **1200**, and the image sensor module (not illustrated) is provided at the rear of the lens module **1200**.

In addition, a baffle **1300** may be provided at the rear of the lens module **1200** in the housing **1010** to block unnecessary light that may flow toward the image sensor, to reduce flare. In the drawing, only one baffle **1300** is illustrated, but two or more baffles **1300** may also be provided.

To implement a telephoto camera, the focal length may be increased, and accordingly, the distance between the lens module **1200** and the image sensor may be increased.

Accordingly, the baffle **1300** may be provided to block unnecessary light in an internal optical path of the housing **1010**. The baffle **1300** is a member fitted into the inner space of the housing **1010**, and may reduce the size of the optical path so that excessive reflection does not occur when light passes through the inner space of the housing **1010**.

The reflection module **1100** and the lens module **1200** are sequentially provided from one side to the other side in the housing **1010**.

For example, as illustrated in the drawing, the housing **1010** may be integrally provided so that both the reflection module **1100** and the lens module **1200** are inserted into the inner space thereof.

By using the integrated housing **1010**, there is no need to separately align the optical axes of the reflection module **1100** and the lens module **1200**, and since the housing is provided as one, the number of components may be reduced, and thus assembly may be very easy.

However, examples of the present disclosure are not limited thereto, and for example, separate housings into which the reflection module **1100** and the lens module **1200** are inserted respectively may be interconnected.

In this embodiment, the reflective member **1150** provided in the reflection module **1100** may have a rotation angle of approximately ± 10 degrees with respect to the first axis (X-axis) (a total of 20 degrees may be rotated), and a rotation angle of approximately ± 25 degrees with respect to the

second axis (Y-axis) (a total of 50 degrees may be rotated), and for example, may rotate in a fairly large range.

Accordingly, a space in the housing **1010** in which the reflection module **1100** is provided may have a greater width, for example, a greater length in the X-axis direction, than that of a space in which the lens module **1200** is provided.

For example, in the housing **1010**, the width of the space in which the reflection module **1100** is provided (length in the X-axis direction) is referred to as 'A', and the width of the space in which the lens module **1200** is provided (the length in the X-axis direction) is referred to 'B', the relationship of ' $2>A/B>1$ ' may be satisfied.

The light whose path is changed by the reflection module **1100** is incident on the lens module **1200**. Therefore, a plurality of lenses provided in the lens module **1200** are provided to be stacked in a Z-axis direction, which is a direction in which light is emitted from the reflection module **1100**.

In addition, the lens module **1200** includes a third driving unit to implement autofocus (AF) and zoom functions.

The lens module **1200** includes a lens holder **1220** provided in a second space **1090** of the housing **1010** and including a stacked lens therein, and a third driving unit for moving the lens holder **1220**.

The lens holder **1220** accommodates a plurality of lenses for imaging a subject, and the plurality of lenses are mounted in the lens holder **1220** along an optical axis. The lens holder **1220** may separately include a lens barrel in which a plurality of lenses are stacked, and a carrier surrounding the lens barrel. Alternatively, a plurality of lenses may be stacked in the lens holder **1220** itself.

The lens holder **1220** is configured to move in the direction of the optical axis (Z axis) to implement an autofocus or zoom function.

The third driving unit generates driving force so that the lens holder **1220** may move in the optical axis (Z axis) direction. For example, the third driving unit may move the lens holder **1220** to change the distance between the lens holder **1220** and the reflection module **1100**.

For example, the third driving unit includes a plurality of third magnets **1241a** and **1243a** and a plurality of third coils **1241b** and **1243b** disposed to face the plurality of third magnets **1241a** and **1243a**.

When power is applied to the plurality of third coils **1241b** and **1243b**, the lens holder **1220** on which the plurality of third magnets **1241a** and **1243a** are mounted may be moved in the direction of the optical axis (Z axis) by an electromagnetic influence between the plurality of third magnets **1241a** and **1243a** and the plurality of third coils **1241b** and **1243b**.

The plurality of third magnets **1241a** and **1243a** are mounted in the lens holder **1220**. For example, the plurality of third magnets **1241a** and **1243a** may be mounted on the side of the lens holder **1220**.

The plurality of third coils **1241b** and **1243b** are mounted in the housing **1010**. For example, a main substrate **1070** may be mounted in the housing **1010** in a state in which the plurality of third coils **1241b** and **1243b** are mounted on the main substrate **1070**.

In this case, for convenience of explanation, in the drawing, both a coil for the reflection module **1100** and a coil for the lens module **1200** are mounted on the main substrate **1070**, but the configuration is not limited thereto, and the main substrate **1070** may be provided as separate substrates on which the coil for the reflection module **1100** and the coil for the lens module **1200** are mounted respectively.

In this embodiment, when the lens holder **1220** is moved, a closed loop control method in which the position of the lens holder **1220** is sensed and fed back is used. Therefore, a third position detection sensor **1243c** is required for closed loop control. The third position detection sensor **1243c** may be a Hall sensor.

The third position detection sensor **1243c** is disposed inside or outside of at least one third coil **1243b**, and the third position detection sensor **1243c** may be mounted on the main substrate **1070** on which the third coil **1243b** is mounted.

The lens holder **1220** is provided in the housing **1010** to be movable in the optical axis (Z axis) direction. For example, a plurality of third ball members **1250** are disposed between the lens holder **1220** and the housing **1010**.

The plurality of third ball members **1250** serve as bearings that guide the movement of the lens holder **1220**, and also function to maintain a gap between the lens holder **1220** and the housing **1010**.

The plurality of third ball members **1250** are configured to roll or slide in the optical axis (Z axis) direction when driving force is generated in the optical axis (Z axis) direction. Accordingly, the plurality of third ball members **1250** guide the movement of the lens holder **1220** in the optical axis (Z axis) direction.

A plurality of seventh guide portions **1221** and **1231** accommodating the plurality of third ball members **1250** are formed on at least one of surfaces of the lens holder **1220** and the housing **1010**, facing each other.

The plurality of third ball members **1250** are accommodated in the plurality of seventh guide portions **1221** and **1231** and are fitted between the lens holder **1220** and the housing **1010**.

The plurality of seventh guide portions **1221** and **1231** may have a shape having a length in the optical axis (Z axis) direction.

The plurality of third ball members **1250** are restricted in movement in the X-axis and Y-axis directions perpendicular to the optical axis in the state of being accommodated in the plurality of seventh guide portions **1221**, **1231** and only move in the optical axis (Z-axis) direction. For example, the plurality of third ball members **1250** may only roll in the optical axis (Z axis) direction.

To this end, the plurality of seventh guide portions **1221**, **1231** may respectively be formed in a shape elongated in the optical axis (Z axis) direction. In addition, cross-sections of the plurality of seventh guide portions **1221** may have various shapes such as a curved shape or a polygonal shape.

In this case, the lens holder **1220** is pressed toward the housing **1010** so that the plurality of third ball members **1250** may maintain contact thereof with the lens holder **1220** and the housing **1010**.

To this end, a third yoke **1260** may be mounted on the bottom surface of the housing **1010** to face the plurality of third magnets **1241a** and **1243a** mounted in the lens holder **1220**. The third yoke **1260** may be a magnetic material.

Attractive force acts between the third yoke **1260** and the plurality of third magnets **1241a** and **1243a**. Accordingly, the lens holder **1220** may be moved in the optical axis (Z axis) direction by the driving force of the third driving unit in a state in contact with the plurality of third ball members **1250**.

The lens holder **1220** is supported by the housing **1010** by the attractive force of the third yoke **1260** and the plurality of third magnets **1241a** and **1243a**, but accordingly, the lens holder **1220** may be separated by an external force such as

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external impacts or the like to collide with other members such as the cover 1030 or the like.

Accordingly, in an example of the present disclosure, a third auxiliary member 1280 may be provided to prevent the lens holder 1220 from being displaced and to absorb an impact even in the case in which shaking by an external force occurs.

The third auxiliary member 1280 serves as a stopper or damper, is provided in an approximately 'E' shape, and both ends thereof may be fitted and fixed to the housing 1010 to cover the lens holder 1220 from the top.

The third auxiliary member 1280 may additionally include a damping member formed of an elastic material on various portions thereof to absorb shock. For example, in the process of moving the lens holder 1220 in the optical axis direction, the front end or the rear end of the lens holder 1220 in the optical axis direction may contact the third auxiliary member 1280, and in this case, both ends of the third auxiliary member 1280 may be provided with a damper 1283.

The third auxiliary member 1280 may be provided as two third auxiliary members to be installed on both sides of the lens holder 1220, respectively.

The housing 1010 is covered by the cover 1030.

The cover 1030 has an opening 1031 so that light is incident therein, and the light incident through the opening 1031 is changed in a traveling direction by the reflection module 1100 so that the light enters the lens module 1200. The cover 1030 may be integrally provided to cover the entire housing 1010, or may be divided into separate members covering the reflection module 1100 and the lens module 1200, respectively.

The opening 1031 provided in the cover 1030 may be provided in a substantially hexagonal shape. The reflection module 1100, according to the present embodiment, is capable of relatively large rotation based on the second axis (Y axis), and accordingly, when the reflective member 1150 is rotated to the maximum, based on the second axis (Y-axis), a short edge of the reflective member 1150 and the side of the cover 1030 may be disposed substantially parallel to each other.

Referring further to FIG. 7, the housing 1010 includes the reflection module 1100 and the lens module 1200 in an internal space. Accordingly, the internal space of the housing 1010 may be divided into a first space 1080 in which the reflection module 1100 is disposed and a second space 1090 in which the lens module 1200 is disposed.

The first space 1080 in which the reflection module 1100 is disposed in the housing 1010 may be provided so that the inner space has a rounded shape, such that a carrier 1110 rotating relatively widely may be easily rotated.

Further, since the plurality of coils 1114, 1134, 1241b, and 1243b are provided in the housing 1010 while being mounted on the main substrate 1070, a plurality of through-holes 1010a, 1010b, 1010c, and 1010d may be provided in the housing 1010 in such a manner that the plurality of coils 1114, 1134, 1241b, and 1243b are exposed to the inner space of the housing 1010.

FIG. 9 is an exploded perspective view of a housing and a reflection module of a camera module according to one or more examples, and FIG. 10A is a detailed upper exploded perspective view of a housing and a reflection module of a camera module according to one or more examples, and FIG. 10B is a detailed lower exploded perspective view of a housing and a reflection module of a camera module according to one or more examples.

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Referring further to FIGS. 9 to 10B, the reflection module 1100 includes the carrier 1110 provided in the housing 1010 and a rotation holder 1130 provided in the carrier 1110.

The carrier 1110 rotates with respect to the housing 1010 about a second axis A2 (an axis parallel to the Y axis), and the rotation holder 1130 rotates with respect to the carrier 1110 about a first axis A1 (an axis parallel to the X axis).

Referring further to FIGS. 14 to 19, the carrier 1110 is provided in the first space 1080 of the housing 1010. The carrier 1110 is closely supported with a first ball member 1111 sandwiched between the bottom surface of the housing 1010 and the carrier 1110, and the carrier 1110 is rotationally driven by a first driving unit. At least a portion of the first space 1080 may be provided in a rounded shape to facilitate a relatively large rotation of the carrier 1110. In more detail, at least a portion of the inner surface of the first space 1080 may be provided to correspond to an arc shape of a circle centered on the second axis A2.

In addition, the carrier 1110 may be provided such that at least a portion thereof has a rounded shape, to facilitate the relatively large rotation thereof in the first space 1080. In further detail, at least a portion of the carrier 1110 may be provided corresponding to the arc shape of a circle centered on the second axis A2.

Further, the rotation holder 1130 mounted on the carrier 1110 may be provided such that at least a portion thereof has a rounded shape, to facilitate the relatively large rotation in the first space 1080. More specifically, at least a portion of the rotation holder 1130 may be provided to correspond to the arc shape of a circle centered on the second axis A2.

The first driving unit includes a first magnet 1113 and a first coil 1114.

Accordingly, a first yoke 1112 is provided on the bottom surface of the housing 1010, and the first yoke 1112 enables the carrier 1110 to be in close contact with the bottom surface of the housing 1010 by attraction force with the first magnet 1113 provided in the carrier 1110.

At least three first ball members (1111) 1111a, 1111b, and 1111c may be provided between the bottom surface of the housing 1010 and the carrier 1110.

One of the three first ball members, a rotating shaft ball 1111a, forms the second axis A2 (an axis parallel to the Y-axis), which is a rotation axis in which the carrier 1110 rotates with respect to the housing 1010, and the other ball members, guide balls 1111b and 1111c may help to facilitate rotation of the carrier 1110.

In this case, the second axis A2 may be perpendicular to a plane including a triangle connecting the three first ball members (1111) 1111a, 1111b, and 1111c.

Since the rotating shaft ball 1111a should form a rotating shaft, the position is not changed and the rotating shaft ball may be rotated or fixed in place by itself while being fixed in one place. Accordingly, the carrier 1110 may rotate about the rotating shaft ball 1111a.

The guide balls 1111b and 1111c are provided on a position other than the rotation axis, to guide the rotation of the carrier 1110, and thus may be provided to roll or slide. Accordingly, the movement of the carrier 1110 may be guided by the rolling or sliding of the guide balls 1111b and 1111c.

Accordingly, a guide portion or the like into which the rotating shaft ball 1111a is inserted is provided on the bottom surface of the housing 1010 and the lower surface of the carrier 1110.

A first guide portion 1121a may be provided in the housing 1010 such that the rotating shaft ball 1111a is inserted, and a second guide portion 1121b may be provided

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in the carrier **1110**. Since the rotating shaft ball **1111a** having a spherical shape should not be moved in position, at least one of the first guide portion **1121a** and the second guide portion **1121b** may be supported by at least three points with the rotating shaft ball **1111a**.

For example, the first guide portion **1121a** and the second guide portion **1121b** may be provided in a shape in which each corner is cut in a triangular pyramid (tetrahedral) shape as illustrated in FIGS. **16A** and **16B**, which will be described later.

Alternatively, the rotating shaft ball **1111a** may be fixedly provided to one of the housing **1010** or the carrier **1110**, and the guide portion that is not moved in position is provided in the other of the housing **1010** or the carrier **1110**.

Third guide portions **1123a** may be provided in the housing **1010** and fourth guide portions **1123b** may be provided in the carrier **1110**, such that the guide balls **1111b** and **1111c** are inserted. Since the spherical guide balls **1111b** and **1111c** are preferably moved in position, the third guide portion **1123a** and the fourth guide portion **1123b** may be provided to be elongated in the rotation direction of the carrier **1110**.

In this embodiment, the third guide portion **1123a** and the fourth guide portion **1123b** may be provided in a straight line shape in the rotation direction.

In the case in which the third guide portion **1123a** and the fourth guide portion **1123b** are provided in a straight line shape, one of the third guide portion **1123a** and the fourth guide portion **1123b** may be provided to have an additional degree of freedom. This is because the guide balls **1111b** and **1111c** support the rotating carrier **1110**, and thus, if configured to be only moved in a straight direction, the guide may not be properly performed.

For example, among the third guide portion **1123a** and the fourth guide portion **1123b**, as illustrated in FIGS. **17A** and **17B**, respectively, selectively, one is supported by at least two points, including a side surface of a 'V-shaped' or 'U-shaped' groove, by the guide balls **1111b** and **1111c**, and the other is supported by one point on a substantially flat bottom surface by the guide ball **1111b**, **1111c**, not contacting the side surface of the groove.

Accordingly, there is no restraint on the side surface inside of the guide portion that does not contact the side of the groove and is supported only by one point on the bottom surface, and thus, the ball may move left and right along the bottom surface, thereby providing an additional degree of freedom. Therefore, even when the carrier **1110** rotates, the rolling of the guide balls **1111b** and **1111c** may be relatively smooth.

Referring to FIG. **17A**, a rotating portion **1110** which rotates relative to a fixed portion **1010** with respect to a rotation axis may be guided by the guide balls **1111b** and **1111c**. The guide balls **1111b** and **1111c** may include a third guide portion **1123a** and a fourth guide portion **1123b** that are provided in a straight line substantially in a rotational direction. In this case, the third guide portion **1123a** provided in the housing that is the fixed portion **1010** has a bottom surface **26a** and both side surfaces **25a**, and the guide balls **1111b** and **1111c** may be supported by at least two points on both side surfaces **25a** (may also be supported up to the bottom surface **26a**).

In addition, the fourth guide portion **1123b** provided in the carrier, which is the rotating portion **1110**, has a bottom surface **26b** and both side surfaces **25b**, and the guide balls **1111b** and **1111c** are by one point on the bottom surface **26a**. When the guide balls **1111b** and **1111c** come into contact with one of the side surfaces **25b**, the guide balls **1111b** and

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1111c are no longer able to roll over, and thus, both side surfaces **25b** may serve as stoppers.

In this manner, the guide balls **1111b** and **1111c** are supported by at least two points in the third guide portion **1123a** provided in the fixed portion **1010** to move linearly along a predetermined path of the guide portion, and are supported by one point on the bottom surface in the fourth guide portion **1123b** provided in the rotating portion **1110**. The guide balls **1111b** and **1111c** supported by one point form a curved moving path on the bottom surface **26b** of the guide portion of the rotating portion **1110** according to the movement of the carrier, which is the rotating portion **1110**.

In addition, referring to FIG. **17B**, the rotating portion **1110** which rotates relative to the fixed portion **1010** with respect to the rotation axis may be guided by the guide balls **1111b** and **1111c**. The guide balls **1111b** and **1111c** may include a third guide portion **1123a-1** and a fourth guide portion **1123b-1**, provided in a straight line in the rotation direction.

In this case, the third guide portion **1123a-1** provided in the housing, which is the fixed portion **1010**, has a bottom surface **26a-1** and both side surfaces **25a-1**, and the guide balls **1111b** and **1111c** are supported by one point on the bottom surface **26a-1**. When the guide balls **1111b** and **1111c** come into contact with either of both side surfaces **25a-1**, the guide balls **1111b** and **1111c** cannot rolling move any more, and thus, both side surfaces **25a-1** may function as stoppers. In addition, the fourth guide portion **1123b-1** provided in the carrier, which is the rotating portion **1110**, has a bottom surface **26b-1** and both side surfaces **25b-1**, and the guide balls **1111b** and **1111c** may be supported by at least two points on both side surfaces **25b-1** (may also be supported up to the bottom surface **26b-1**).

In this manner, the guide balls **1111b** and **1111c** are supported by at least two points in the fourth guide portion **1123b-1** provided in the rotating portion **1110** to move linearly along a predetermined path of the guide portion, and are supported by one point on the bottom surface in the third guide portion **1123a-1** provided in the fixed portion **1010**, and the one point supported guide balls **1111b** and **1111c** form a curved path on the bottom surface **26a-1** of the guide portion of the fixed portion **1010** according to the movement of the carrier, which is the rotating portion **1110**.

Referring further to FIGS. **18** to **20**, a first coil **1114** is provided on the bottom of the housing **1010**, and a first magnet **1113** facing the first coil **1114** is provided on the carrier **1110**. In addition, to detect the rotation position of the carrier **1110**, a first position detection sensor **1115** may be provided in the housing **1010** to face the first magnet **1113**.

The first magnet **1113** may be provided in a rounded shape in consideration of the rotation of the carrier **1110**. The first magnet **1113** may have an inner end and an outer end in a circular arc shape, and in more detail, the first magnet **1113** has an inner end and an outer end of a circular arc centered on the rotating shaft ball **1111a**.

For example, the first magnet **1113** may be provided in a shape in which a donut is partially cut off. In addition, the first magnet **1113** may be provided to have an N pole and an S pole in the rotation direction.

In the carrier **1110**, a rear yoke **1113a** that maintains or further improves the performance of the first magnet **1113** by focusing magnetism on the rear surface of the first magnet **1113**, for example, between the carrier **1110** and the first magnet **1113**, may be further provided. In consideration of the fact that the first magnet **1113** has a rounded shape, the rear yoke **1113a** may be provided in a relatively larger shape to face the first magnet **1113**.

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In addition, the first coil **1114** may also be disposed in a position corresponding to the first magnet **1113**. The first coil **1114** may be provided as one or two or more first coils, and the first coil **1114** may be disposed in a rounded shape or a bent shape corresponding to the shape of the first magnet **1113**. For example, when two first coils **1114** are provided, the coils may be disposed to implement a bent shape as a whole, for example, a 'V' shape.

For example, when the first magnet **1113** is magnetized with three poles of 'N pole, S pole, N pole' or 'S pole, N pole, S pole' in the rotation direction of the carrier **1110**, the first coil **1114** may be provided in two so that respective coils may be disposed to face the poles at the middle and left and right ends at the same time.

For example, in the case of the 'N pole, S pole, N pole' magnet, one of the two first coils **1114** may be disposed to substantially face the N pole on the left and the left half of the S pole in the middle, and the other may be disposed to substantially face the N pole on the right and the right half of the S pole in the middle.

Of course, the first magnet **1113** may be provided as two magnets each separated by two poles, and may be disposed to face the two first coils **1114**, respectively. In this case, the first magnet **1113** may have a rounded shape or a straight shape.

On the other hand, the first magnet **1113** may be provided between the first ball members (**1111**) **1111a**, **1111b**, and **1111c**. In more detail, the first magnet **1113** may be disposed between the rotating shaft ball **1111a** and the guide balls **1111b** and **1111c**.

The first ball members (**1111**) **1111a**, **1111b**, and **1111c**, for example, one rotating shaft ball **1111a** and two guide balls **1111b** and **1111c**, may be disposed in a triangular shape.

The center of gravity or the geometric center of the first magnet **1113** may be provided inside of a triangle formed by the first ball members (**1111**) **1111a**, **1111b**, and **1111c**. The carrier **1110** is in close contact with the housing **1010** by the attraction between the first yoke **1112** and the first magnet **1113**, and the carrier **1110** is not inclined to either side by the attraction generated at this time.

On the other hand, since the center of gravity or the geometric center of the first magnet **1113** is provided inside of the triangle formed by the first ball members (**1111**) **1111a**, **1111b**, and **1111c**, when the power supply to the reflection module **1100** is stopped, the first carrier **1110** may be moved to the initial position by the attraction of the magnet **1113** and the first yoke **1112**.

The initial position may be adjusted depending on the arrangement of the first magnet **1113** and the first yoke **1112**, for example, in the case of the present embodiment, the position in which the reflective member **1150** is aligned parallel to the optical axis direction may be preferable.

The two first coils **1114** facing the first magnet **1113** may also be disposed between the rotating shaft ball **1111a** and the guide balls **1111b** and **1111c**.

In addition, the first position detection sensor **1115** for sensing the position of the carrier **1110** may be disposed to face the first magnet **1113**. The first position detection sensor **1115** may be a Hall sensor. One or two or more first position detection sensors **1115** may be provided for more accurate position detection of the carrier **1110**.

The first position detection sensor **1115** may be provided between the first coil **1114** and the rotating shaft ball **1111a** when viewed in a plan view. In the case of the rotating carrier **1110**, the moving distance thereof becomes longer as the distance from the rotating shaft **1111a** increases, and when moving away from the rotating shaft ball **1111a**

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forming the rotating shaft by considering that the carrier **1110** of this embodiment rotates a large amount (rotation angle approximately ± 25 degrees), since the movement distance increases, a relatively large number of position detection sensors may be required.

Accordingly, in the present embodiment, the first position detection sensor **1115** may be disposed in an inside of the first coil **1114**, for example, on a position close to the rotating shaft ball **1111a** forming a rotating shaft.

On the other hand, a first yoke **1112** may be provided in the housing **1010** to face the first magnet **1113**. The first yoke **1112** may serve as a pulling yoke that closely contacts the carrier **1110** to the housing **1010**.

In addition, the first yoke **1112** may be provided in the housing **1010** to surround (finishing) the first coil **1114**, and accordingly, magnetic field (magnetic) leakage of the first magnet **1113** or the first coil **1114** may be prevented.

The first yoke **1112** may enable the carrier **1110** to be in close contact with the housing **1010** by the attraction with the first magnet **1113**, and thus, the first yoke **1112** may be provided to face the first magnet **1113** with the coil **1114** or the like therebetween, in a shape similar to the first magnet **1113**.

For example, the first yoke **1112** may be provided in a rounded shape in which a donut is partially cut, and may be provided to be larger than both the first magnet **1113** and the first coil **1114** to surround the same.

With additional reference to FIGS. **11** to **13**, the rotation holder **1130** is provided in the carrier **1110**. The rotation holder **1130** is provided with the reflective member **1150**, and the rotation holder **1130** is rotationally driven by a second driving unit. The second driving unit includes a second magnet **1133** and a second coil **1134**.

The reflective member **1150** may change the traveling direction of light. For example, the reflective member **1150** may be a mirror or a prism reflecting light (for convenience of explanation, the reflective member **1150** is illustrated as a prism in the drawings related to an embodiment).

The reflective member **1150** is fixed to the rotation holder **1130**. The rotation holder **1130** is provided with a mounting surface **1136** on which the reflective member **1150** is mounted.

The mounting surface **1136** of the rotation holder **1130** may be configured as an inclined surface such that the path of light is changed. For example, the mounting surface **1136** may be an inclined surface inclined 30 to 60 degrees with respect to the optical axis (Z axis) of the plurality of lenses. The inclined surface of the rotation holder **1130** may face the opening **1031** of the cover **1030** through which light is incident.

Further, on an end of the mounting surface **1136** in the lens module (**1200**) direction in the optical axis direction, a plurality of protrusions **1136a** may be provided toward the reflective member **1150** to reduce the occurrence of flares due to light reflection, diffraction, or the like.

The end of the protrusion **1136a** may be sharply formed, and the protrusion **1136a** may be provided over a predetermined area of the end portion of the mounting surface **1136**.

The rotation holder **1130** may be supported in close contact with the carrier **1110** with two second ball members **1131** sandwiched therebetween.

Accordingly, the carrier **1110** and the rotation holder **1130** are selectively provided with a first magnetic body **1138** and a second magnetic body **1132**, respectively, and the rotation holder **1130** is in close contact with the carrier **1110** by the attraction of the first magnetic body **1138** and the second magnetic body **1132**.

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When the first magnetic body **1138** is a magnet, for example, the fourth magnet **1138**, a rear yoke is further provided on the rear surface of the fourth magnet **1138** to maintain or further improve the performance of the fourth magnet **1138** by focusing magnetism.

As illustrated in detail in FIG. 20, when the fourth magnet **1138** is provided in the carrier **1110**, the rear yoke **1113a** may be used in common. For example, the carrier **1110** is provided with the rear yoke **1113a** that maintains or further improves the performance of the first magnet **1113** by focusing the magnetism of the first magnet **1113**, and the rear yoke **1113a** may be provided to cover the rear surface of the fourth magnet **1138** by increasing the length of the rear yoke **1113a**. Accordingly, the rear yoke **1113a** may include an extension portion **1113b** extending to the rear surface of the fourth magnet **1138**.

In this case, in the case of the reflection module **1100** (folded module) according to one or more examples, a first magnetic body **1138** and a second magnetic body **1132** may be provided selectively on the carrier **1110** and the rotation holder **1130**. Further, the rotation holder **1130** may be provided to be supported by the carrier **1110** by the attractive force of the first magnetic body **1138** and the second magnetic body **1132**. The first magnetic body **1138** and the second magnetic body **1132** may be provided to face each other in the second axis A2 direction. The reflection module **1100** (folded module) according to the present embodiment has a structure in which the rotation holder **1130** is placed on an upper portion of the carrier **1110**, and thus, the rotation holder **1130** may be supported toward the carrier **1110** for driving stability.

In this case, the first magnetic body **1138** or the second magnetic body **1132** is a magnetic material, and may be a material having magnetism, for example, a material that is magnetized in a magnetic field (including both metal or non-metal materials). The first magnetic body **1138** or the second magnetic body **1132** may be a pulling magnet or a pulling yoke.

For example, when the first magnetic body **1138** is a pulling magnet, the second magnetic body **1132** may be a pulling yoke or a pulling magnet. In addition, when the first magnetic body **1138** is a pulling yoke, the second magnetic body **1132** may be a pulling magnet.

The rotation holder **1130** rotates relative to the carrier **1110** based on a first axis (an axis parallel to the X axis) connecting the two second ball members **1131** to each other.

Accordingly, the carrier **1110** is provided with two first support portions **1141** on both sides in the X-axis direction, and the rotation holder **1130** is provided with two second support portions **1143** which are mounted on the first support portions **1141** and are disposed on both sides in the X-axis direction to correspond to each other.

The second ball member **1131** may be provided between the pairs of the first support portions **1141** and the second support portions **1143** provided on both sides, respectively. In addition, the two second ball members **1131** may form a first axis A1 that is parallel to the X-axis direction and is a rotation axis of the rotation holder **1130**.

Since the two second ball members **1131** need to form a rotating shaft, the positions thereof are not changed and may be rotated or fixed in place by themselves while being fixed in one place. Accordingly, the rotation holder **1130** may rotate about the first axis A1 formed by the two second ball members **1131** as an axis.

Accordingly, the first support portion **1141** and the second support portion **1143** are provided with a guide portion (for

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example, a groove, a notch, a collar, a divot, etc.) into which the second ball member **1131** is inserted.

The first support portion **1141** of the carrier **1110** is provided with a fifth guide portion **1141a**, and the second support portion **1143** of the rotation holder **1130** is provided with a sixth guide portion **1143a**, such that the second ball members **1131** are inserted.

Since the spherical second ball member **1131** should not be moved in position, at least one of the fifth guide portion **1141a** and the sixth guide portion **1143a** may be supported by at least three points by the second ball member **1131**. In this case, the rotating shaft ball **1111a** may be provided to be fixed to either the housing **1010** or the carrier **1110**, and a guide portion that is not moved in position may be provided on the other of the housing **1010** or the carrier **1110**.

For example, FIGS. 16A and 16B are reference diagrams illustrating an example of a structure in which a ball member of a camera module, according to one or more examples, is fixed to be supported by three points, on a guide portion.

Referring to FIGS. 16A and 16B, since the ball members **1111a** and **1131** forming the rotating shaft cannot move their position, the positions thereof may be fixed by a three-point support structure.

The ball members **1111a** and **1131** may be inserted into the guide portions **1121a**, **1121b**, **1141a**, and **1143a**.

In addition, the ball member inserted into the guide portion may maintain a supported state by contacting the guide portion only at three points P to maintain an accurate position inside the guide portion.

If the ball member contacts the guide portion at four or more points, it may be driven in a biased state, such as forming contact at only three points according to manufacturing tolerances or driving conditions of the guide portion or the ball member.

To this end, the guide portions **1121a**, **1121b**, **1141a**, and **1143a** may be provided in a shape in which each corner is cut in a triangular pyramid (tetrahedron) shape.

The spherical ball members **1111a** and **1131** are supported at three points P on the inner side surfaces of the guide portions **1121a**, **1121b**, **1141a**, and **1143a**, and thus, the guide portions include three first surfaces **21**. Therefore, the three contact points P of the ball members **1111a** and **1131** and the guide portions **1121a**, **1121b**, **1141a** and **1143a** are formed on the first surfaces **21**.

In this case, the first surface **21** is a portion of the side surface, and the side surface includes the first surface **21** on which the ball members **1111a** and **1131** contact, and a second surface **23** provided between the first surfaces **21** (for example, provided adjacent to two of the first surfaces) and not in contact with the ball members **1111a** and **1131**.

In addition, a triangular pyramid (tetrahedron) may be implemented by extending the three side surfaces (first surfaces **21**) to which the ball members **1111a** and **1131** are respectively in point contact. For example, a line segment formed by extending the three side surfaces that are in point contact with the ball members **1111a** and **1131** and intersect each other may implement the corners of a triangular pyramid (tetrahedron). In addition, a triangular pyramid implemented by extending three side surfaces may be an equilateral triangular pyramid.

On the other hand, the guide portions **1121a**, **1121b**, **1141a**, and **1143a** may be provided in a shape in which each vertex is cut in a triangular pyramid (tetrahedron) shape.

In the triangular pyramid, the incision on the inner apex of the guide portion may form a bottom **10** of the guide portion (**1121a**, **1121b**, **1141a**, **1143a**), and the incised portions of the remaining three apexes of the guide portion

on the entrance side may form the second surface **23** that does not contact the ball members **1111a** and **1131**, among the side surfaces.

Both the bottom **10** and the second surface **23** are formed by cutting off the vertices of the triangular pyramid, and thus, may all have a triangular shape, and the ball members **1111a** and **1131** do not contact the bottom **10** and the second surface **23**. In addition, the entrances of the guide portions **1121a**, **1121b**, **1141a**, and **1143a** may have a hexagonal shape because all the vertices are cut from the bottom of the triangular shape of a triangular pyramid (tetrahedron).

On the other hand, the bottoms of the guide portions **1121a**, **1121b**, **1141a**, and **1143a** may have a triangular shape.

A second coil **1134** is provided on a side surface of the housing **1010**, and a second magnet **1133** facing the second coil **1134** is provided on the rotation holder **1130**. In addition, to detect the rotation position of the rotation holder **1130**, the housing **1010** may be provided with a second position detection sensor **1135** to face the second magnet **1133**.

The second magnet **1133** may be magnetized to have an N pole and an S pole in a direction of the second axis **A2** perpendicular to the first axis **A1**, and the rotation holder **1130** may rotate with respect to the carrier **1110** based on the first axis **A1** by interaction between the second magnet **1133** and the second coil **1134**.

In this case, the side surface of the housing **1010** on which the second coil **1134** is provided may mean a side surface perpendicular to the optical axis.

The rotation holder **1130** may have a rounded end portion in consideration of being provided on the carrier **1110** that rotates. In addition, the second magnet **1133** may be provided on the end of a rounded shape of the rotation holder **1130**, which is an opposite side to the direction in which the reflective member **1150** is installed. Accordingly, the second magnet **1133** may also be provided in a rounded shape. The second magnet **1133** may have an inner end and an outer end in a circular arc shape, and in more detail, the second magnet **1133** has an inner end and an outer end to correspond to an arc shape of a circle centered on the rotating shaft ball **1111a**.

On the rotation holder **1130**, a rear yoke **1133a** may further be provided to maintain or further improve the performance of the second magnet **1133** by focusing magnetism on the rear surface of the second magnet **1133**, for example, between the rotation holder **1130** and the second magnet **1133**. Considering that the second magnet **1133** has a rounded shape, the rear yoke **1133a** may have a relatively larger shape corresponding to the second magnet **1133**.

The second coil **1134** may also be disposed in a position corresponding to the second magnet **1133**.

The second coil **1134** may be provided as one or two or more second coils, and the second coil **1134** may be disposed in a rounded shape or a bent shape to correspond to the shape of the second magnet **1133**. For example, the second coil **1134** may be provided in two, and the two coils **1134** may be disposed on both sides to face the second magnet **1133**, and when the second coil **1134** is provided in two, the second coils may be disposed to implement an overall bent shape, for example, a 'V' shape.

Of course, the second magnet **1133** may be provided as two separate magnets, and may also be disposed to face the two first coils **1134**, respectively. In this case, the second magnet **1133** may have a rounded shape or a linear shape.

The second magnet **1133** may be provided at an end that is farthest from the first axis **A1**, which is a rotation axis of

the rotation holder **1130**. For example, the second magnet **1133** may be provided on the end of the rounded shape.

The second yoke **1132** (second magnetic body) and the fourth magnet **1138** (first magnetic body) serve as a pulling yoke and a pulling magnet, respectively, and may be optionally provided on the rotation holder **1130** or the carrier **1110**. The second yoke **1132** and the fourth magnet **1138** may be respectively provided in one or two or more to face each other in the Y-axis direction.

Since the rotation holder **1130** is supported by the carrier **1110** and rotates relative thereto, the second yoke **1132** and the fourth magnet **1138** disposed to face each other may be disposed in positions not to interfere with the rotation of the rotation holder **1130**.

Accordingly, in this embodiment, the second yoke **1132** and the fourth magnet **1138** are very close to the rotating shaft **A1** and respectively two thereof may be provided below the first axis **A1**, to sufficiently exert pulling force and not to interfere with the rotation of the rotation holder **1130** as much as possible. Accordingly, the first axis **A1**, the second yoke **1132**, and the fourth magnet **1138** may be aligned in the Y-axis direction, and may be provided on approximately the same position in the optical axis (Z-axis) direction (see FIG. 21).

The second position detection sensor **1135** for detecting the position of the rotation holder **1130** may be disposed to face the second magnet **1133**. The second position detection sensor **1135** may be a Hall sensor. The second position detection sensor **1135** may be provided in one or two or more for more accurate position detection of the rotation holder **1130**.

The second position detecting sensor **1135** may be provided vertically between the two second coils **1134**.

On the other hand, the carrier **1110** may be provided with two first support portions **1141** protruding on both sides in the X-axis direction, and the rotation holder **1130** may be provided with two second support portions **1143** mounted on the first support portions **1141**, on both sides in the X-axial direction.

In this case, the first support portion **1141** may have an open upper portion, and a structure in which the second support portion **1143** is fitted to the first support portion **1141** from the top to the lower portion may be provided.

In addition, the carrier **1110** may further include a first auxiliary member **1160** covering the open upper portion of the first support portion **1141**. Since the second support portion **1143** rotates, the first auxiliary member **1160** may not be in close contact with the second support portion **1143** and may have a slight gap therewith, so as not to interfere with the rotation of the second support portion **1143**.

The first auxiliary member **1160** may serve as a stopper that prevents the rotation holder **1130** from being separated from the carrier **1110** or a buffer member that absorbs the shock when the second support portion **1143** collides with other parts due to displacement.

When the second support portion **1143** is coupled to the first support portion **1141**, the first auxiliary member **1160** may be fitted into the carrier **1110** through the side surface to cover the second support portion **1143** from the top. Accordingly, a slit-shaped coupling hole **1142** may be provided in the carrier **1110** such that the first auxiliary member **1160** is inserted.

The first auxiliary member **1160** may be provided with a bent portion **1163** for preventing a separation thereof on an end of a body **1161** having a 'E' shape for firmed coupling. In addition, the body **1161** may include a fixed portion **1161a** of which a portion is fitted to the carrier **1110**, and a damping

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portion **1161b** disposed on the second support portion **1143** and coupled to a damping member **1165**.

The damping member **1165** has a damping protrusion **1165a** protruding toward the second support portion **1143**, and the damping protrusion **1165a** may be provided to face the second ball member **1131** provided on the second support portion **1143**. The structure of the damping protrusion **1165a** may efficiently perform a buffering or stopping role of the rotation holder **1130**.

In addition, the damping member **1165** may further include a buffer protrusion **1165b** protruding toward the cover **1030**.

Further, in the present embodiment, when the rotation holder **1130** is separated from the carrier **1110**, the end of the rotation holder **1130**, for example, the upper portion of the rounded portion on which the second magnet **1133** is installed may also collide with an inner wall of the cover **1030**.

Accordingly, in the present embodiment, a second auxiliary member **1040** may be provided to prevent the end of the rotation holder **1130** from colliding with the cover **1030** or to absorb an impact.

The second auxiliary member **1040** may be provided in a 'E' shape such that an end portion thereof may be fitted to the side surface of the housing **1010** to be fixed. In addition, the second auxiliary member **1040** may be provided with a buffer member **1043** that is further provided between the second auxiliary member and the rotation holder **1130** or the cover **1030** to facilitate shock absorption.

The second ball member **1131** may be provided between the first support portion **1141** and the second support portion **1143**. In addition, the two second ball members **1131** provided on both sides may form a first axis **A1** that is formed as a connection line therebetween which is in parallel with the X-axis direction while being the rotation axis of the rotation holder **1130**.

FIG. **21** is a reference diagram for explaining a positional relationship between two axes in which a reflection module rotates in a camera module, according to one or more examples.

Referring to FIG. **21**, the reflection module **1100** according to one or more examples may be provided in such a manner that the reflective member **1150** rotates about two axes.

The carrier **1110** provided in the housing **1010** may rotate based on the first axis **A1** formed by the rotating shaft ball **1111a**, and the rotation holder **1130** provided in the carrier **1110** may rotate based on the second axis **A2** formed by the second ball member **1131**.

Accordingly, the reflective member **1150** provided in the rotation holder **1130** may rotate based on the first axis **A1** and the second axis **A2**. In the illustration of FIG. **21**, a portion illustrated by a dotted line includes the rotation holder **1130**, and the rotation holder **1130** may rotate with respect to the first axis **A1**. In addition, a portion illustrated by a solid line includes the carrier **1110**, and the carrier **1110** may rotate based on the second axis **A2**.

The first axis **A1** and the second axis **A2** intersect each other, and the rotating shaft ball **1111a** forming the second axis **A2** and the two ball members **1131** forming the first axis **A1** may be disposed on a plane on which both the first axis **A1** and the second axis **A2** are provided. In detail, when viewed in the direction in which light is incident, for example, in the direction of the second axis **A2**, the rotation axis ball **1111a** and the two ball members **1131** may be aligned in the direction of the first axis **A1** perpendicular to the optical axis.

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In the reflection module **1100** according to the present embodiment, the intersection of the first axis **A1** and the second axis **A2** may be formed at approximately the center of the mounting surface **1136** on which the reflective member **1150** is mounted. For example, the first axis **A1** may be formed along the mounting surface **1136** in a direction parallel to the X-axis direction, and the second axis **A2** may be provided to penetrate approximately through the center of the mounting surface **1136**.

The reflective member **1150** is mounted on the mounting surface **1136**, and the approximately center of the mounting surface **1136** may substantially correspond to the center of the reflective surface. Therefore, when the intersection point of the first axis **A1** and the second axis **A2** is formed at approximately the center of the mounting surface **1136**, the actual rotation amount of the rotation holder **1130** approximately matches the rotation amount of the reflective surface, for example, the mounting surface **1136**, and thus, controlling shake correction or tracking may be relatively very easy.

FIGS. **22A** and **22B** are views for explaining a shape of an opening of a cover (cover member) in a camera module, according to one or more examples.

The cover **1030** of the present embodiment includes an opening **1031** through which light is incident, and the opening **1031** may be provided in a substantially hexagonal shape.

The carrier **1110** of the reflection module **1100** rotates based on a second axis **A2** parallel to the direction (Y-axis direction) in which light is incident on the camera module **1000**, and the rotation angle thereof is approximately ± 25 degrees as a relatively very large degree.

Therefore, the opening **1031** is provided in a substantially hexagonal shape to sufficiently receive the incident light while reducing the incidence of unnecessary light.

For example, when the carrier **1110** rotates about the second axis **A2** with respect to the housing **1010**, the reflective member **1150** rotates, and the angle may change slightly in a state in which the edge portion is aligned in parallel with the optical axis direction (Z axis direction). Accordingly, in consideration of a maximum rotation angle of the reflective member **1150**, a side of the entrance hole positioned in the lateral direction of the housing **1010** may be formed substantially parallel to the edge of the reflective member **1150**.

FIG. **22A** illustrates a shape when the reflective member **1150** is rotated at a maximum in a clockwise direction with respect to the second axis **A2**, and FIG. **22B** illustrates a shape in which the reflective member **1150** is rotated at a maximum counterclockwise with respect to the second axis **A2**.

FIG. **23** is a perspective view of an integrated substrate installed in the camera module, according to one or more examples, and FIG. **24** is a perspective view illustrating a state in which the integrated substrate is installed in the housing of the camera module, according to one or more examples.

The main substrate **1070** according one or more examples may be provided integrally. Further, the main substrate **1070** may be provided with coils **1114** and **1134** for driving the reflection module **1100** of the first and second driving units, and a plurality of coils **1241b** and **1243b** for driving the lens module **1200** of the third driving unit, which may be mounted on an inner surface of the main substrate. In addition, components (not illustrated) such as active devices and various passive devices, gyro sensors (not illustrated),

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and the like may be mounted on the outer surface of the main substrate. Accordingly, the main substrate **1070** may be a double-sided substrate.

In addition, the main substrate **1070** may be coupled to the housing **1010**. Since the integrally provided main substrate **1070** is coupled to the housing **1010** in the state in which all the coils of the driving unit are mounted on the main substrate **1070**, assembly may be relatively easy.

FIG. **25** is a reference diagram illustrating two lens barrels of a camera module according to one or more other examples, and FIG. **26** is a reference diagram illustrating three lens barrels of a camera module according to one or more still other examples.

Referring to FIG. **25**, a camera module **1001** according to one or more other examples may include a reflection module **1100** and a lens module **1201**.

In addition, the lens module **1201** may include two or more lens holders **1221** and **1223** to implement an improved autofocus or zoom function.

The two or more lens holders **1221** and **1223** may respectively move in the direction of the optical axis (Z axis), and may be individually controlled.

When the camera module **1001** includes two or more lens holders **1221** and **1223**, one thereof may be used for autofocus, and the remaining or all lens holders may be used to implement the zoom function. Therefore, further improved autofocus or zoom function may be implemented.

Referring to FIG. **26**, a camera module **1002** according to one or more still other examples may include a reflection module **1100** and a lens module **1202**.

In addition, the lens module **1202** may include three or more lens holders **1231**, **1233** and **1235** to implement an improved autofocus or zoom function.

Any one of the three or more lens holders **1231**, **1233** and **1235**, for example, the lens holder **1231** provided closest to the reflection module **1100**, has a fixed position, and the remaining two or more lens holders **1233** and **1235** may respectively move in the direction of the optical axis (Z axis), and may be individually controlled.

The camera module **1002** may be provided in such a manner that a portion **1231** of the lenses aligned in the optical axis direction is fixed, and two or more remaining lens holders **1233** and **1235** may move in the optical axis direction.

When a portion of the plurality of lenses is additionally provided as a fixed lens, a further improved optical effect may be implemented. In addition, since any one of the movable lens holders may be used for autofocus, and the remaining or all lens holders may be used to implement the zoom function, further improved autofocus or zoom function may be implemented.

Through these examples, the camera module and the portable electronic device including the camera module according to one or more examples may implement functions such as autofocus, zoom, shake correction, PIP, tracking, etc., while the structure is relatively simple and the driving may be significantly easy.

As set forth above, according to one or more examples, shaking may be easily corrected in not only imaging a fixed subject but also in capturing video of a moving subject.

In addition, a folded module (a camera module) capable of tracking a moving subject and a portable electronic device including the same may be provided.

While specific examples have been shown and described above, it will be apparent after an understanding of this disclosure that various changes in form and details may be

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made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

What is claimed is:

1. A folded module comprising:

a housing comprising a first space and a second space; a reflection module disposed in the first space and configured to be rotatable about a first axis perpendicular to an optical axis and a second axis perpendicular to the optical axis and the first axis;

a lens module disposed in the second space along the optical axis; and

a driving unit comprising a plurality of magnets and a plurality of coils facing the plurality of magnets, the plurality of magnets and the plurality of coils being configured to drive the reflection module and the lens module,

wherein the plurality of magnets comprise:

a first magnet having a flat shape having an inner surface and an outer surface each being parallel to the second axis and having an arc shape of a circle centered on the second axis, and

a second magnet having a curved shape having an arc shape of a circle centered on the second axis.

2. The folded module of claim 1, wherein the first space and the second space satisfy the following relationship:

$$1 < A/B < 2$$

where A is a length of the first space in a direction parallel to the first axis, and B is a length of the second space in a direction parallel to the first axis.

3. The folded module of claim 1, wherein at least a portion of the first space has a rounded shape corresponding to an arc shape of a circle centered on the second axis perpendicular to the optical axis and the first axis.

4. The folded module of claim 3, wherein the reflection module comprises:

a carrier disposed in the first space of the housing; and a rotation holder disposed in the carrier and configured to hold a reflective member,

wherein each of a portion of the carrier and a portion of the rotation holder has a rounded shape corresponding to an arc shape of a circle centered on the second axis perpendicular to the optical axis and the first axis.

5. The folded module of claim 1, wherein the reflection module comprises:

a carrier disposed in the first space of the housing; and a rotation holder disposed in the carrier and configured to hold a reflective member,

wherein the carrier comprises a first support portion provided on opposite sides of the carrier in a direction parallel to the first axis,

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the rotation holder comprises a second support portion provided on opposite sides of the rotation holder in a direction parallel to the first axis and disposed on the first support portion, and

the reflection module further comprises a first auxiliary member coupled to the first support portion and covering the second support portion so that there is a gap between the second support portion and the first auxiliary member in a direction parallel to the second axis.

6. The folded module of claim 5, wherein the first auxiliary member comprises a damping protrusion protruding toward the second support portion in the direction parallel to the second axis.

7. The folded module of claim 5, wherein the first support portion comprises a slit-shaped coupling hole into which the first auxiliary member is inserted.

8. The folded module of claim 5, further comprising a second auxiliary member coupled to the housing and covering at least a portion of the rotation holder.

9. The folded module of claim 5, further comprising two ball members disposed between the first support portion and the second support portion and forming the first axis.

10. The folded module of claim 1, wherein the lens module comprises a lens holder in which a plurality of lenses are mounted along the optical axis.

11. The folded module of claim 10, further comprising a third auxiliary member coupled to the housing and covering at least a portion of the lens holder.

12. The folded module of claim 1, further comprising a cover covering the housing,

wherein the cover comprises an opening having a hexagonal shape through which light is incident onto the reflection module.

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13. The folded module of claim 1, further comprising: an image sensor module disposed on the optical axis so that the lens module is disposed between the reflection module and the image sensor module, and

one or more baffles is disposed between the lens module and the image sensor module.

14. The folded module of claim 1, further comprising a main substrate on which at least some of the plurality of coils are disposed,

wherein the main substrate is mounted on the housing so that at least a portion of the main substrate is mounted on a side surface of the housing and is inclined relative to the optical axis.

15. The folded module of claim 14, wherein the portion of the main substrate that is mounted on the side surface of the housing and is inclined relative to the optical axis faces the second magnet.

16. The folded module of claim 1, wherein a maximum rotation angle of the reflection module about the second axis is more than twice a maximum rotation angle of the reflection module about the first axis.

17. The folded module of claim 1, wherein a length of the first space in a direction parallel to the first axis is greater than a length of the second space in a direction parallel to the first axis.

18. The folded module of claim 1, wherein a portion of the housing comprising the first space protrudes farther in a direction parallel to the first axis than a portion of the housing comprising the second space.

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